Structure and Predictability in Foreign Exchange Markets: A Koopman-Operator and Graph-Embedded Dynamic Mode Decomposition Framework

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MSc Dissertation submitted to the University of Surrey

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Abstract

Foreign exchange (FX) markets are exceptionally difficult to forecast due to heavy-tailed returns, volatility clustering, rapid regime shifts and constantly evolving cross-asset relationships. This thesis investigates whether a Koopman-operator approach, implemented through Graph-Embedded Dynamic Mode Decomposition (GEDMD), can uncover short-horizon structure within these conditions.

Daily close prices for five major USD currency pairs (EUR/USD, USD/JPY, GBP/USD, USD/CNH and USD/CAD) were collected via the OANDA API and used to develop a complete GEDMD forecasting pipeline tailored to FX dynamics. The method combines Hankel time-delay embeddings constructed from normalised price trajectories, correlation-confidence graphs to identify statistically meaningful cross-asset neighbours, and Tikhonov-regularised local operator estimation. Operators are learned in rolling windows and assessed through reconstruction diagnostics, eigenvalue-modulus stability checks, Monte Carlo perturbations and extensive hyperparameter search.

Three modelling scenarios are analysed. The first reproduces the original GEDMD configuration proposed for equity markets and performs poorly in FX due to unstable graph structure and misaligned regularisation. The second introduces an FX-specific hyperparameter search, substantially improving stability and directional forecasting performance. The third incorporates a scaled, data-adaptive Tikhonov penalty, yielding the most coherent operators, the strongest cross-sectional ranking ability and the highest economic performance across thresholds.

Although absolute point forecasts remain noisy—consistent with the near-martingale nature of FX—GEDMD consistently extracts low-dimensional signals with cross-sectional predictive value. Directional accuracy improves with thresholding, calibration curves show monotonic structure and ranking-based portfolios achieve higher Sharpe ratios than an equal-weight benchmark.

The analysis also highlights inherent limitations. Linear operators degrade during large macroeconomic shocks, neighbourhood graphs can reorganise abruptly during regime shifts, and model stability weakens in strongly nonlinear or ill-conditioned environments. These limitations reflect structural features of FX markets rather than flaws in the methodology. Overall, the findings show that a Koopman-based, graph-embedded operator-learning framework can produce stable, interpretable and economically useful forecasts when appropriately adapted to FX dynamics. This thesis presents the first comprehensive application of GEDMD to foreign exchange markets.

Keywords: Koopman operator theory; Dynamic Mode Decomposition; Graph-Embedded DMD; FX forecasting; operator learning; delay embeddings; correlation-confidence graphs; Tikhonov regularisation.

Acknowledgements

I would like to express my deepest gratitude to my supervisor, **Dr. Dorje Brody**, whose guidance, clarity of thought and intellectual generosity shaped every stage of this thesis. His insights into dynamical systems, operator methods and quantitative modelling were invaluable, and his encouragement consistently pushed the project toward greater mathematical and empirical rigour.

My sincere thanks also go to the academic staff of the programme, whose teaching provided the theoretical foundations that made this research possible. The modules in mathematics, statistics, financial data science and machine learning created the intellectual environment in which this project could be conceived and executed. I am also thankful to my classmates for the conversations, debates and shared problem-solving that contributed indirectly to the formulation of ideas throughout this work.

I am indebted to my family and close friends for their understanding and steady support during the long months of development and revision. Their belief in my work—and in my ability to complete it—played an essential role in bringing this thesis to completion.

Any remaining errors or oversights are entirely my own.

Contents

1 Introduction **1**

2 Literature Review **4**

2.1 Koopman Operator Theory 4

2.2 Dynamic Mode Decomposition and Operator Learning 5

2.3 Graph-Based Extensions and GEDMD 5

2.4 Stylised Facts and Microstructure of FX Markets 6

2.5 Gaps in the Literature and the Absence of DMD in FX 6

2.6 Why Standard DMD Struggles in FX and Why GEDMD Is More Suitable . . . 7

3 Theoretical Framework **8**

3.1 Koopman Operator Foundations 8

3.2 Dynamic Mode Decomposition as a Koopman Approximation 9

3.3 Time-Delay Embeddings 9

3.4 Graph-Embedded Local Operators 10

3.5 Tikhonov-Regularised Operator Estimation 10

3.5.1 Justification for Scaled Regularisation 11

3.6 Rolling-Window Approximation 13

3.7 From Operator Estimates to Forecasts 13

4 Methodology **14**

4.1 Data Preparation 14

4.2 Time-Delay Embedding 15

4.3 Correlation-Confidence Graphs 15

4.4 Composite Observables 16
4.5 Tikhonov-Regularised Operator Estimation 16
4.6 Forecasting Procedure 17
4.7 Rolling-Window Framework 17
4.8 Monte Carlo Robustness Analysis 17
4.9 Evaluation of Forecasts and Trading Performance 18

5 Foreign Exchange Data Architecture **19**
5.1 Data Source and Structure 19
5.2 Temporal Alignment and Cleaning 20
5.3 Normalisation for Koopman Estimation 21
5.4 Return Representation 21
5.5 Statistical Behaviour of FX Series 21

6 Results **24**
6.1 Operator Reconstruction and Stability 24
6.2 Monte Carlo Robustness Analysis 25
6.3 Forecast Accuracy 25
6.4 Directional Structure and Precision-Recall Behaviour 27
6.5 Calibration Properties 28
6.6 Portfolio Performance 29
6.7 Hyperparameter Behaviour 33
6.8 Observed Limitations 34

7 Discussion and Directions for Future Research **35**
7.1 Why Equity Parameters Fail in FX: A Domain Transfer Analysis 35
7.2 Economic Sources of GEDMD Predictability 36
7.3 Discussion 37
7.4 Directions for Future Research 38

8 Conclusion **40**

A Appendix A: Data Collection and Preprocessing **44**

B Appendix B: Graph Construction Diagnostics **47**

C Appendix C: Local Operator Stability Diagnostics **50**

C.1 Singular Value Structure of Snapshot Matrices 50

C.2 Role of Singular Values in Regularisation 51

C.3 Eigenvalue Diagnostics of Local Operators 52

C.4 Interpretation 53

D Appendix D: Monte Carlo Uncertainty Analysis **54**

E Appendix E: Walk-Forward Validation Procedure **58**

F Appendix F: Hyperparameter Search and Sensitivity Analysis **61**

G Appendix G: Portfolio Construction Framework **64**

H Appendix H: Computational Framework Overview **70**

1 Introduction

Foreign exchange (FX) markets constitute the most liquid financial markets globally, with daily turnover exceeding $7.5 trillion according to the Bank for International Settlements' Triennial Central Bank Survey *Triennial Central Bank Survey: Foreign Exchange Turnover in April 2022* 2022. Their continuous trading cycle, sensitivity to global macroeconomic conditions and deep liquidity make FX a central component of international finance. These same features create substantial modelling challenges: returns exhibit heavy tails, volatility clustering and abrupt structural breaks Cont, 2001; Andersen et al., 2003, while cross-asset dependencies reorganise rapidly in response to monetary policy decisions, risk sentiment and global liquidity cycles Lyons, 2001; Engel, 2016. Short-horizon FX returns are often modelled as approximately martingale, implying limited predictability. From a dynamical-systems perspective, however, it remains an open question whether transient, low-dimensional patterns exist over short windows that can be detected without imposing restrictive parametric assumptions.

This thesis investigates that question through an operator-theoretic lens. Specifically, it evaluates whether Graph-Embedded Dynamic Mode Decomposition (GEDMD), an extension of Dynamic Mode Decomposition (DMD) grounded in Koopman operator theory Koopman, 1931; Mezić, 2013, can be adapted in a mathematically coherent and empirically meaningful way to FX markets. Standard DMD constructs a global linear operator that advances observables through time Schmid, 2010; Rowley et al., 2009, while GEDMD extends this framework by embedding temporal memory via delay coordinates Tu et al., 2014 built from normalised price trajectories and incorporating cross-asset structure using correlation-confidence neighbourhood graphs Fisher, 1915; Yip et al., 2023. The resulting method estimates a local Koopman operator for each asset based on its own embedding and those of its statistically significant neighbours.

Although GEDMD performs well in equity markets Mann and Kutz, 2016; Yip et al., 2023, its applicability to FX is not established. Structural differences between the two asset classes challenge the transferability of operator-learning methods. Equities exhibit persistent sectoral clusters and smoother dependence structures, which support stable DMD-type approximations. FX markets, in contrast, display heavier tails Cont, 2001, faster-moving cross-sectional networks Lyons, 2001, and sharp regime transitions Engel, 2016. These properties undermine the assumptions required for standard DMD to remain stable—contributing to the absence of DMD-based research in FX—while creating uncertainty about whether GEDMD can accommodate such volatility and structural change. Accordingly, this thesis frames the applicability of GEDMD

to FX as an empirical question.

The empirical analysis proceeds in three stages, designed to evaluate both the feasibility and limitations of GEDMD in FX forecasting. In Scenario 1, the full architecture of Yip et al. Yip et al., 2023 is implemented exactly as proposed, including their embedding length, sampling window, neighbourhood size and regularisation scheme. Applied to five major USD-centric currency pairs (EUR/USD, USD/JPY, GBP/USD, USD/CNH and USD/CAD), this equity-optimised configuration underperforms a naïve equal-weight benchmark. The result highlights a fundamental limitation: hyperparameter regimes calibrated for equities do not transfer directly to FX, where temporal scales, noise properties and correlation dynamics differ markedly.

Scenario 2 addresses this gap by conducting a comprehensive hyperparameter search over embedding dimensions, neighbourhood sizes, Fisher-z significance thresholds, sampling windows and regularisation strengths. The optimal configuration differs substantially from the original equity-based design, demonstrating stronger ranking behaviour, improved precision–recall characteristics and significantly higher Sharpe ratios. This stage shows that GEDMD can be made empirically effective in FX, but only when its components are tuned to the statistical and economic properties of currency markets.

Scenario 3 refines the regularisation mechanism by introducing a scaled Tikhonov penalty in which the regularisation strength is proportional to the square of the leading singular value of the design matrix. Motivated by FX's heavy-tailed noise structure Cont, 2001 and the instability of ill-conditioned ridge inverses Tikhonov, 1963, this modification yields better-conditioned operators, lower reconstruction error and improved perturbation robustness. Under this configuration, GEDMD achieves its strongest empirical performance, producing forecasts with stable predicted–realised correlations and economically meaningful returns when incorporated into ranking- and threshold-based portfolio rules Mann and Kutz, 2016; Lyons, 2001.

Robustness is assessed through Monte Carlo perturbation analysis, eigenvalue-modulus diagnostics, operator-stability metrics and strict walk-forward validation. Across these tests, the scaled-Tikhonov implementation exhibits stable behaviour in regimes characterised by persistent cross-asset structure but degrades during episodes of abrupt correlation breakdown—a limitation consistent with both operator-theoretic constraints and FX microstructure.

The contributions of this thesis are fourfold. First, it provides one of the first systematic evaluptions of GEDMD in FX markets, identifying both its potential and its structural limitations. Second, it demonstrates that operator-learning models are highly sensitive to hyperparameter regimes and that parameter transferability across asset classes cannot be assumed. Third, it introduces a scaled Tikhonov regularisation scheme that materially improves operator stability and forecasting performance in FX. Fourth, it establishes a unified operator-theoretic framework—spanning forecasting, performance evaluation, robustness analysis and walk-forward validation—for assessing Koopman-based methods in complex, nonlinear financial systems.

The remainder of this document is organised as follows. Chapter 2 reviews the literature

on Koopman operator theory, DMD, GEDMD and FX market structure. Chapter 3 develops the theoretical framework, including the Koopman perspective, graph-embedded operators and associated research questions. Chapter 4 describes the methodological design, including embeddings, graph construction, regularisation choices and forecasting architecture. Chapter 5 outlines the data and preprocessing steps. Chapter 6 reports the empirical results across all scenarios. Chapter 7 provides an integrated discussion of the findings, including domain-transfer analysis, economic interpretation, limitations and directions for future research. Chapter 8 concludes.

Mathematics Subject Classification (2020): Primary: 37M10 (Time series analysis), 37C60 (Non-autonomous dynamical systems), 91G60 (Numerical methods in finance). Secondary: 37A30 (Ergodic theory), 65F22 (Ill-posed problems), 62M10 (Time series).

2 Literature Review

Forecasting foreign exchange (FX) markets requires methods capable of handling heavy-tailed behaviour, regime shifts and time-varying dependence structures. Linear operator-learning approaches offer a compelling analytical framework, but none of the assumptions that sustain them in physical or equity-market settings transfer automatically to FX. This chapter therefore reviews the theoretical and empirical foundations of Koopman operator methods, Dynamic Mode Decomposition (DMD), graph-embedded variants and FX stylised facts, with particular attention to the tension between their underlying assumptions and the realities of FX dynamics.

2.1 Koopman Operator Theory

Koopman operator theory reframes nonlinear evolution in terms of a linear, infinite-dimensional operator acting on observables. For a discrete-time system $x_{t+1} = F(x_t)$, the Koopman operator $\mathcal{K}$ acts through

$$(\mathcal{K}g)(x) = g(F(x)),$$

providing a linear representation of the evolution of observables even when F is nonlinear. The classical formulation originates with Koopman and von Neumann Koopman, 1931, while modern developments connect Koopman spectral properties to ergodic theory, coherent structures and data-driven decomposition Mezić, 2013.

Finite-dimensional approximations of $\mathcal{K}$ can capture meaningful short-horizon dynamics if the chosen observables approximate an invariant or slowly changing subspace S. L. Brunton, B. W. Brunton, and Kutz, 2016; S. L. Brunton and Kutz, 2019. In systems with coherent, low-dimensional structure—such as fluid flows—this approximation is effective. In FX, however, it is not known whether similar structure persists over relevant horizons. Thus, the role of Koopman theory in this context is exploratory: it provides a lens for discovering whether any approximately linear evolution exists locally, rather than assuming this structure a priori.

2.2 Dynamic Mode Decomposition and Operator Learning

Dynamic Mode Decomposition (DMD) provides a practical, data-driven approximation to Koopman dynamics. Given snapshot matrices

$$X_1 = [x_1, x_2, \ldots, x_m], \qquad X_2 = [x_2, x_3, \ldots, x_{m+1}],$$

DMD estimates a linear operator A satisfying

$$X_2 \approx AX_1.$$

Originally introduced in the study of fluid flows Schmid, 2010; Rowley et al., 2009, DMD identifies coherent temporal-spatial modes whose eigenvalues reflect growth, decay and oscillation. Delay-embedded variants Tu et al., 2014 enrich the observable space by incorporating temporal memory through lagged snapshots of the underlying price series.

Applications to financial markets Mann and Kutz, 2016 show that DMD can extract short-lived predictive structure in equities. Yet fundamental differences between equities and FX complicate direct transfer. FX returns display heavy tails, heteroskedasticity and abrupt structural breaks Cont, 2001, and cross-asset relationships evolve rapidly across regimes. These features create ill-conditioned regression matrices and weaken the persistence needed for stable operator estimation. DMD therefore functions in FX as a hypothesis-testing instrument: it can reveal structure if such structure exists, but it does not assume its presence.

2.3 Graph-Based Extensions and GEDMD

Graph-based operator-learning acknowledges that multivariate financial systems are not homogeneous collections of independent assets. Sectoral relationships, contagion channels and network interactions shape financial co-movements Billio et al., 2012. Graph-regularised Koopman approximations Williams, Kevrekidis, and Rowley, 2015 exploit this structure by embedding neighbourhood information directly into operator estimation.

Graph-Embedded Dynamic Mode Decomposition (GEDMD) Yip et al., 2023 extends DMD in two key ways. First, it uses delay embeddings to construct temporally enriched observables from normalised price trajectories. Second, it selects each asset's neighbourhood using Fisher-z-filtered correlation confidence intervals Fisher, 1915. A local operator is estimated for each asset on the composite observable space formed by its own embedding and those of its statistically significant neighbours. This architecture performs well in equity markets, where correlations often reflect persistent sectoral organisation.

FX markets, however, differ fundamentally. Correlation networks reorganise rapidly in response to macroeconomic cycles, interest rate differentials and global risk conditions Andersen et al., 2003. Additional algebraic constraints—triangular arbitrage, USD dominance and cross-currency symmetry—further complicate the structure. Whether GEDMD's neighbour-selection mechanism and local operators can remain stable under such conditions is not demonstrated in prior work. Its applicability to FX is therefore empirical rather than presumed.

2.4 Stylised Facts and Microstructure of FX Markets

Any operator-learning method applied to FX must contend with its well-documented empirical regularities. FX returns exhibit excess kurtosis, heavy tails and asymmetry Cont, 2001, alongside volatility clustering driven by macroeconomic announcements, policy shifts and geopolitical events Andersen et al., 2003. These features create noisy, rapidly shifting landscapes in which linear approximations may only hold locally, if at all.

Cross-sectional dependencies are equally unstable. USD-centric pairs anchor the global FX system, while triangular arbitrage relations impose algebraic consistency constraints Lyons, 2001. Empirical studies show that correlations vary substantially across regimes and often shift abruptly during stress periods Engel, 2016. Neighbourhood selection in this context is nontrivial and sensitive to window length, noise and transient effects.

Microstructure research reveals further complexity. Order flow, dealer inventory management and liquidity provision shape short-horizon movements Evans and Lyons, 2002. These mechanisms generate transient predictability but also introduce rapid state changes that challenge any fixed linear approximation.

Taken together, the stylised facts do not rule out the presence of exploitable structure; they simply constrain its nature. If FX markets contain predictable components, they are likely to be local, transient and embedded in patterns of cross-asset adjustment. GEDMD, with its local operators and graph-based construction, is therefore a candidate method—but its effectiveness depends entirely on empirical performance rather than theoretical inevitability.

2.5 Gaps in the Literature and the Absence of DMD in FX

A notable gap in the existing literature is the absence of DMD or Koopman-based operator-learning methods applied to FX markets. The most prominent financial application—Mann and Kutz Mann and Kutz, 2016—focuses exclusively on equities and offers no discussion of why FX was omitted. This absence is unlikely to be accidental. Structural properties of FX pose challenges to the core assumptions that make DMD successful in physical and equity contexts.

Equity markets exhibit more persistent dependence structures, clearer sectoral clustering and relatively stable autocorrelation patterns, enabling coherent mode extraction. In contrast, FX markets are characterised by heavier tails Cont, 2001, sharp volatility bursts Andersen et al., 2003, and rapidly evolving correlation networks driven by global macroeconomic cycles Lyons, 2001; Engel, 2016. These conditions undermine the stability of the DMD regression matrix and weaken the persistence of coherent modes.

The scarcity of FX-focused DMD research therefore reflects not an oversight, but deep structural obstacles. Whether any Koopman-based approximation can remain stable under FX-specific noise, heteroskedasticity and cross-asset dynamics is an empirical question rather than an assumed extension. This thesis addresses precisely that gap by testing GEDMD under conditions known to challenge operator-learning methods.

2.6 Why Standard DMD Struggles in FX and Why GEDMD Is More Suitable

Standard DMD assumes that the underlying system exhibits coherent, slowly changing structures and that snapshot pairs arise from approximately linear evolution Schmid, 2010; Rowley et al., 2009. FX violates these assumptions in multiple ways.

First, volatility bursts flatten the singular-value spectrum of the data matrix, making the implicit inversion in DMD extremely sensitive to noise. Second, modes are not persistent: FX dependence structures reorganise rapidly across regimes, precluding stable eigenstructure estimation. Third, standard DMD lacks mechanisms for incorporating cross-asset constraints such as triangular arbitrage, USD dominance and macroeconomic co-movements Lyons, 2001.

GEDMD addresses these weaknesses directly. Graph-embedded neighbourhoods restrict estimation to statistically meaningful and economically interpretable relationships, reducing noise and preventing spurious cross-asset coupling. Local operators replace the single global model, enabling stability even when the global correlation structure reorganises. Delay embeddings capture the short-lived memory effects arising from microstructure dynamics Evans and Lyons, 2002. Together, these features make GEDMD a more robust and empirically justifiable extension of DMD for FX markets.

3 Theoretical Framework

This chapter unifies the operator-theoretic foundations, structural considerations and modelling principles that motivate the approach developed in this study. The intention is not to assume *a priori* that a Koopman-based method must succeed in FX markets, but to lay out the mathematical and conceptual tools that make such an approach worth investigating. The chapter therefore brings together the linearisation perspective of Koopman operator theory, the finite-dimensional approximations implemented through Dynamic Mode Decomposition (DMD), the structural refinements introduced by graph-embedded variants, and the statistical considerations needed to construct stable operators from noisy, non-stationary FX data.

3.1 Koopman Operator Foundations

Consider a discrete-time nonlinear dynamical system

$$x_{t+1} = F(x_t),$$

with state $x_t \in \mathbb{R}^d$. Traditional modelling approaches attempt to specify or estimate the nonlinear map F, an endeavour often infeasible in financial settings where interactions are high-dimensional, shocks are frequent and structural regimes shift unpredictably. Koopman operator theory instead shifts the focus from state evolution to the evolution of observables. For any scalar observable $g : \mathbb{R}^d \to \mathbb{R}$, the Koopman operator $\mathcal{K}$ acts as

$$(\mathcal{K}g)(x) = g(F(x)).$$

Although F may be nonlinear, the operator $\mathcal{K}$ is *linear* in the space of observables. This linearity enables the use of spectral tools to describe nonlinear dynamics, including long-run tendencies encoded in eigenfunctions and eigenvalues Koopman, 1931; Mezić, 2013. If one identifies or approximates a finite-dimensional invariant subspace of observables, then the evolution of the system within that subspace can be represented by a finite-dimensional linear operator. This observation motivates the construction of numerical approximations to $\mathcal{K}$ directly from data S. L. Brunton and Kutz, 2019.

Financial systems exhibit transient structure, intermittent regimes and nonlinear interactions.

The Koopman viewpoint does not claim that markets are governed by stable, low-dimensional operators; rather, it offers a principled way to approximate short-term dynamics without specifying a parametric stochastic model, making it a natural candidate for exploratory analysis in FX markets.

3.2 Dynamic Mode Decomposition as a Koopman Approximation

Dynamic Mode Decomposition (DMD) provides a numerical method for constructing a finite-dimensional approximation of the Koopman operator. Given snapshots

$$X_1 = \begin{bmatrix} x_1 & x_2 & \cdots & x_m \end{bmatrix}, \qquad X_2 = \begin{bmatrix} x_2 & x_3 & \cdots & x_{m+1} \end{bmatrix},$$

DMD seeks a matrix A such that

$$X_2 \approx AX_1.$$

When computed via least squares using the singular value decomposition (SVD), A represents the best-fit linear operator advancing the observable state by one time step Schmid, 2010; Rowley et al., 2009. Under appropriate conditions, the eigenvalues of A approximate those of the Koopman operator, and its modes describe coherent temporal structures Tu et al., 2014.

However, financial data rarely satisfy the assumptions under which a single global linear operator provides a stable approximation. Heavy tails, heteroskedasticity, sudden structural breaks and time-varying dependencies all complicate operator estimation Cont, 2001. These limitations motivate the refinements introduced below.

3.3 Time-Delay Embeddings

To increase the expressiveness of the observable space, the analysis uses time-delay (Hankel) embeddings. For a univariate series $\{p(t)\}$, a delay vector of length L is

$$\mathbf{h}(t) = \Big[p(t-L+1),\, p(t-L+2),\, \ldots,\, p(t)\Big]^\top.$$

Takens' theorem Takens, 1981 shows that delay coordinates can reconstruct the local structure of a nonlinear dynamical system. In practice, delay embeddings provide DMD with richer temporal information by incorporating lagged values of the underlying price series, improving its ability to capture short-term dynamics in systems characterised by persistence, cycles or regime shifts Tu et al., 2014.

For multivariate FX data, embeddings for each currency pair may be concatenated with

those of related assets, forming the basis for graph-embedded operator estimation.

3.4 Graph-Embedded Local Operators

While classical DMD estimates a single operator for the full system, financial assets evolve within structured networks shaped by economic and algebraic constraints. Graph-Embedded Dynamic Mode Decomposition (GEDMD) incorporates this structure by estimating a separate local operator B_i for each asset i. The operator is constructed on an observable subspace determined by a neighbourhood N_i of statistically relevant assets Yip et al., 2023.

Neighbourhood selection uses Fisher-z-transformed confidence intervals of pairwise correlations Fisher, 1915. Given a sample correlation $\hat{\rho}_{ij}$, the transformation

$$z_{ij} = \frac{1}{2} \log\left(\frac{1+\hat{\rho}_{ij}}{1-\hat{\rho}_{ij}}\right)$$

stabilises variance, allowing significance testing. Only neighbours whose lower confidence bounds exceed a threshold are retained. This avoids spurious graph connections in environments, such as FX markets, where dependence structures evolve rapidly.

The composite embedding for asset i is

$$\mathbf{H}_i(t) = \begin{bmatrix} \mathbf{h}_i(t) \\ \mathbf{h}_{j_1}(t) \\ \vdots \\ \mathbf{h}_{j_{|N_i|}}(t) \end{bmatrix}.$$

3.5 Tikhonov-Regularised Operator Estimation

For each asset, GEDMD solves the regularised regression

$$\min_{B_i} \left\{ \left\| X_2^{(i)} - B_i X_1^{(i)} \right\|_F^2 + \lambda \|B_i\|_F^2 \right\},$$

where $X_1^{(i)}$ and $X_2^{(i)}$ contain the composite embeddings at times t and $t+\tau$. If $X_1^{(i)} = USV^\top$, the solution is

$$B_i = X_2^{(i)} V S_\lambda^{-1} U^\top, \qquad S_\lambda^{-1} = \operatorname{diag}\left(\frac{s_k}{s_k^2+\lambda}\right).$$

Regularisation is indispensable in financial applications because FX data exhibit collinearity, heavy tails and rapidly shifting covariance structures Cont, 2001; Andersen et al., 2003. In this thesis, two Tikhonov schemes are considered:

- **Yip-style adaptive regularisation.** Following Yip et al., 2023, the baseline scheme chooses λ adaptively in each window to target a desired condition number for the snapshot matrix. This is the regularisation used in **Scenario 1**, where all other hyperparameters are exactly those reported by Yip et al., 2023 for equities, and in **Scenario 2**, where the same adaptive rule is retained but embedding and graph hyperparameters are tuned for FX.
- **Scaled Tikhonov regularisation.** The alternative scheme introduced in this thesis sets

$$\lambda = \gamma s_1^2,$$

  where s_1 is the leading singular value of $X_1^{(i)}$ and $\gamma > 0$ is a dimensionless hyperparameter Tikhonov, 1963; Rowley et al., 2009; Schmid, 2010. This scaled formulation is used in **Scenario 3**, while all other hyperparameters are held fixed at the FX-optimised values obtained in Scenario 2.

The multi-scenario structure therefore isolates three distinct effects: (i) transplanting the original equity-optimised configuration of Yip et al., 2023 into FX (Scenario 1); (ii) retuning embedding and graph hyperparameters for FX while keeping the Yip-style adaptive regulariser (Scenario 2); and (iii) replacing that adaptive rule with the scaled Tikhonov scheme tailored to heavy-tailed, regime-dependent FX embeddings (Scenario 3).

3.5.1 Justification for Scaled Regularisation

Regularisation is a fundamental component of operator estimation in high-dimensional, noisy or ill-conditioned environments. In the context of Graph-Embedded Dynamic Mode Decomposition (GEDMD), the inverse problem underlying local Koopman operator estimation often becomes unstable because the snapshot matrix $X_1^{(i)}$ exhibits highly variable singular values across rolling windows. These fluctuations arise naturally from FX market characteristics—heavy-tailed innovations, volatility clustering, and rapidly shifting cross-asset covariance structures (Cont, 2001; Andersen et al., 2003). When singular values collapse during turbulent periods, a fixed regularisation parameter λ can either under-penalise or over-penalise the operator estimate, leading to numerical fragility or excessive shrinkage. This motivates a scaled regularisation scheme in which the penalty adapts directly to the conditioning of the inverse problem.

Ill-posedness of the Operator Inversion. The standard Tikhonov-regularised solution to the optimisation

$$\min_{B_i} \|X_2^{(i)} - B_i X_1^{(i)}\|_F^2 + \lambda \|B_i\|_F^2$$

takes the closed-form expression

$$B_i = X_2^{(i)} V S_\lambda^{-1} U^\top, \qquad S_\lambda^{-1} = \operatorname{diag}\left(\frac{s_k}{s_k^2 + \lambda}\right),$$

where $X_1^{(i)} = USV^\top$ is the thin SVD. When some singular values s_k are small, the factors s_k^{-1} inflate substantially, amplifying noise and producing operators that are extremely sensitive to perturbations (Tikhonov, 1963). In FX data, such low-rank behaviour arises naturally during volatility spikes, macroeconomic announcements, and episodes of correlation breakdown, all of which distort the geometry of delay embeddings and compress the singular value spectrum (Cont, 2001). This makes the operator-inversion problem inherently ill-posed.

Limitations of Constant λ Schemes in FX Markets. A constant regularisation parameter λ assumes that the scale and conditioning of $X_1^{(i)}$ remain stable across time windows. This assumption holds relatively well in equity markets, where embedding matrices tend to exhibit smoother singular-value decay and more stable cross-asset correlation structures (Mann and Kutz, 2016). In FX, however:

- singular values vary sharply across regimes, compressing during volatility clusters;
- the neighbourhood size changes dynamically as correlation-confidence graphs reorganise (Engel, 2016);
- heavy-tailed returns induce irregular perturbations leading to abrupt conditioning changes (Cont, 2001).

While Yip et al. (2023) advocate an adaptive Tikhonov scheme that targets a fixed condition number, many practical DMD implementations in finance still treat λ as effectively constant across windows. The analysis in this thesis therefore focuses on the behaviour of such constant-λ baselines in FX and motivates the scaled formulation introduced below.

A fixed λ cannot simultaneously stabilise the operator in stressed windows while preserving information in calm windows. Under-regularisation yields unstable operators with high perturbation sensitivity, whereas excessive regularisation collapses the operator toward a low-energy mapping that loses predictive structure.

Scaling Regularisation by the Leading Singular Value. To address these issues, this thesis adopts a scaled Tikhonov formulation of the form

$$\lambda = \gamma s_1^2,$$

where s_1 is the largest singular value of $X_1^{(i)}$ and γ is a dimensionless hyperparameter. This design draws on classical results in ill-posed inverse problems (Tikhonov, 1963) and modern interpretations of data-driven Koopman operator learning (Schmid, 2010; Rowley et al., 2009). The justification is threefold:

1. **Scale Adaptation:** tying λ to s_1^2 ensures that the penalty scales with the numerical magnitude of the embedding in each window.
2. **Conditioning Stabilisation:** the ratio λ/s_k^2 determines the strength of shrinkage in each singular direction. Scaling maintains stable relative penalisation even when the singular

spectrum shifts dramatically due to regime changes.

3. **Noise Dampening:** since FX embeddings frequently contain extreme movements, scaled λ acts as a volatility-normalised regulariser, suppressing noise amplification during stressed periods without overshrinking in tranquil periods.

Koopman-Theoretic Interpretation. The Koopman operator approximates the linear evolution of a nonlinear dynamical system on an observable subspace (Koopman, 1931; Mezić, 2013). When the structure of this subspace evolves due to regime shifts or changes in cross-asset relationships, the Koopman projection surface also changes. Scaled regularisation ensures that the resulting finite-dimensional approximation:

- respects the geometry of the observable space,
- preserves interpretable dominant modes,
- remains stable across evolving FX regimes.

Empirical Motivation. As shown in Chapter 6 (Results), Scenario 3 (scaled regularisation) produces the most stable operator family, strongest reconstruction behaviour, and highest directional performance. This empirical evidence reinforces the theoretical motivation for adopting a scaled Tikhonov penalty in FX applications.Within the empirical analysis, the scaled scheme is only applied in Scenario 3. Scenarios 1 and 2 instead employ the Yip-style adaptive Tikhonov rule described above, so that Scenario 2 differs from Scenario 1 only through FX-specific choices of embedding and graph parameters, whereas Scenario 3 isolates the incremental contribution of the scaled regulariser itself.

3.6 Rolling-Window Approximation

Financial markets are non-stationary, so operator estimates must adapt over time. A rolling-window procedure updates embeddings, neighbourhoods and operators at each step, ensuring that the approximation reflects the prevailing market regime rather than aggregating incompatible states across periods Andersen et al., 2003; Yip et al., 2023. This yields a sequence of local operators rather than a single static mapping.

3.7 From Operator Estimates to Forecasts

Each local operator B_i yields a forecast of the embedded state for asset i, which is mapped back to predicted *percentage price changes*. Because these operators capture relative dynamics, forecasts tend to be more informative for ranking or threshold-based decisions than for precise point prediction. This motivates the portfolio construction rules analysed later, which rely on directional or ordinal information Mann and Kutz, 2016.

4 Methodology

This chapter details the full empirical framework through which the graph-embedded Koopman operator approach is implemented in foreign exchange (FX) markets. The methodology is shaped by the challenges specific to FX data, including heavy-tailed returns, regime-dependent correlations and structural non-stationarity Cont, 2001; Andersen et al., 2003. No existing work provides a complete blueprint for applying Graph-Embedded Dynamic Mode Decomposition (GEDMD) to FX, and therefore the methodology serves both as an implementation plan and as a systematic evaluation of whether operator-learning can yield coherent short-horizon forecasts in this environment.

The chapter proceeds in stages. It begins with data preparation and normalisation, followed by time-delay embedding of price trajectories, construction of correlation-confidence neighbourhood graphs, and estimation of Tikhonov-regularised local operators. The forecasting procedure, rolling-window design, robustness analyses and economic evaluation metrics complete the empirical pipeline. The implementation is consistent with the theoretical framework presented earlier while extending it to incorporate FX-specific considerations and the multi-scenario structure used later in the thesis.

4.1 Data Preparation

Let $p_i(t)$ denote the closing price of currency pair i at time t. These series are aligned on a common daily grid to avoid distortions caused by asynchronous trading. Missing observations are forward filled, consistent with the microstructure convention that the last quoted price remains valid in the absence of immediate trade Lyons, 2001. The aligned dataset forms a matrix $P \in \mathbb{R}^{T \times N}$, where T is the number of observations and N is the number of currency pairs.

Within each rolling calibration window, prices are normalised by dividing each series by its final value in the window. If the window ends at t^*, the normalised prices are

$$\widetilde{p}_i(t) = \frac{p_i(t)}{p_i(t^*)}.$$

This transformation stabilises the singular value decomposition used in operator estimation by eliminating numerical scale disparities while preserving relative trajectory information. A

detailed description of data collection and preprocessing appears in Appendix A.

4.2 Time-Delay Embedding

To encode short-horizon temporal structure, each normalised series is embedded into an L-dimensional delay-coordinate vector,

$$\mathbf{h}_i(t) = \left[\widetilde{p}_i(t-L+1),\, \widetilde{p}_i(t-L+2),\, \ldots,\, \widetilde{p}_i(t)\right]^\top.$$

The use of delay embeddings follows the principle that temporal memory can be represented geometrically in an expanded observable space, as formalised by Takens Takens, 1981 and operationalised in delay-embedded DMD Tu et al., 2014. FX data often display short-lived persistence and cyclical volatility patterns Andersen et al., 2003, making delay embeddings a natural choice for capturing such behaviour in operator-learning.

The embedding dimension L is treated as a hyperparameter, and its effect on stability and predictability is analysed through the multi-scenario structure described below.

4.3 Correlation-Confidence Graphs

Graph-embedded operator learning requires identifying, for each asset, a set of neighbours whose dynamics contain statistically meaningful information. To avoid the instability of raw correlations in FX markets, neighbour sets are defined using Fisher-z-transformed confidence intervals Fisher, 1915. For simple return series $r_i(t)$ and $r_j(t)$, compute the Fisher statistic

$$z_{ij} = \frac{1}{2}\log\left(\frac{1+\hat{\rho}_{ij}}{1-\hat{\rho}_{ij}}\right),$$

where $\hat{\rho}_{ij}$ is the sample correlation in the current window. Under mild assumptions, z_{ij} is approximately normal with variance $1/(n-3)$. A confidence interval of the form

$$z_{ij} \pm z_{\alpha/2}\sqrt{\frac{1}{n-3}}$$

is then computed. Asset j is included in the neighbourhood N_i when the lower bound of this interval exceeds a statistical threshold. This rule ensures that only robust cross-asset relationships enter the operator, a critical property given the rapid evolution of FX correlation networks Engel, 2016. Detailed diagnostics for graph construction, including significance filtering behaviour and neighbourhood stability, are provided in Appendix B.

Neighbourhoods are updated in every rolling window, permitting the graph to adapt to structural changes such as monetary-policy cycles or global risk shifts.

4.4 Composite Observables

For each asset i, the final observable vector aggregates temporal and cross-asset information. If the neighbours of i are $j_1, \ldots, j_k$, the composite embedding is

$$\mathbf{H}_i(t) = \begin{bmatrix} \mathbf{h}_i(t) \\ \mathbf{h}_{j_1}(t) \\ \vdots \\ \mathbf{h}_{j_k}(t) \end{bmatrix}.$$

This representation implements the structural refinement of GEDMD Yip et al., 2023, but its content is determined empirically through the FX-specific neighbourhoods defined above. The dimensionality of $\mathbf{H}_i(t)$ varies across assets and windows, reflecting the time-varying nature of FX cross-dependencies.

4.5 Tikhonov-Regularised Operator Estimation

Given the composite embeddings, snapshot matrices are formed as

$$X_1^{(i)} = \Big[\mathbf{H}_i(t_1), \ldots, \mathbf{H}_i(t_M)\Big], \qquad X_2^{(i)} = \Big[\mathbf{H}_i(t_1+\tau), \ldots, \mathbf{H}_i(t_M+\tau)\Big].$$

The local Koopman operator B_i is estimated by solving the Tikhonov-regularised least-squares problem

$$\min_{B_i} \left\| X_2^{(i)} - B_i X_1^{(i)} \right\|_F^2 + \lambda \|B_i\|_F^2.$$

If $X_1^{(i)} = USV^\top$ is the thin SVD, the solution is

$$B_i = X_2^{(i)} V S_\lambda^{-1} U^\top, \qquad S_\lambda^{-1} = \operatorname{diag}\left(\frac{s_k}{s_k^2 + \lambda}\right).$$

Regularisation is essential in FX because extreme events and strong collinearity frequently distort the regression matrix, producing unstable or non-informative operators without penalisation Cont, 2001. Two formulations are examined: a fixed-λ design and a scaled variant in which λ is proportional to the square of the leading singular value, following principles of ill-posed problem theory Tikhonov, 1963. Operator stability diagnostics, including singular-value behaviour and eigenvalue-modulus inspection, are presented in Appendix C.

4.6 Forecasting Procedure

Let $\mathbf{x}_i(t)$ denote the composite observable vector at time t. The τ-step forecast is obtained through

$$\widehat{\mathbf{x}}_i(t+\tau) = B_i \mathbf{x}_i(t).$$

The predicted normalised price is then extracted from the corresponding coordinate of the forecasted embedding. Because the operator acts on a price-based observable space, the resulting prediction is mapped back into a *percentage price change* for the underlying asset. These forecasts are used primarily for directional or ranking-based decisions rather than for precise point estimates, consistent with empirical findings in operator-learning applications Mann and Kutz, 2016.

4.7 Rolling-Window Framework

To preserve out-of-sample integrity and adapt to non-stationary dynamics, estimation is embedded within a rolling-window structure. At each iteration, the calibration window advances by one step, delay embeddings are recomputed, neighbourhoods reconstructed and operators re-estimated. This design mirrors standard practice in empirical finance Andersen et al., 2003 and is essential for evaluating whether the operator retains coherence under changing regimes. Implementation details for the walk-forward mechanics are documented in Appendix E.

4.8 Monte Carlo Robustness Analysis

To examine sensitivity to sampling noise and numerical perturbations, the analysis includes Monte Carlo perturbation testing. Perturbed matrices are generated via

$$X_{1,\epsilon}^{(i)} = X_1^{(i)} + \epsilon W_1, \qquad X_{2,\epsilon}^{(i)} = X_2^{(i)} + \epsilon W_2,$$

where W_1 and W_2 contain i.i.d. noise. The corresponding operators $B_{i,\epsilon}$ satisfy

$$\Delta(B_i) = \|B_{i,\epsilon} - B_i\|_F.$$

This diagnostic reveals whether the operator captures genuine local structure or whether the behaviour originates from numerical fragility, an important distinction when working with heavy-tailed data. Full Monte Carlo simulation design, assumptions and perturbation results are reported in Appendix D.

4.9 Evaluation of Forecasts and Trading Performance

Forecasts are assessed using both statistical and economic measures. Statistical evaluation examines the correlation between predicted and realised *simple* returns, calibration patterns and standard error metrics such as MSE and MAE. Economic relevance is evaluated through cumulative returns and annualised Sharpe ratios derived from ranking-based or threshold-based portfolio rules. The Sharpe ratio is computed as

$$\text{Sharpe} = \frac{\sqrt{n_{\text{per year}}}\,\overline{R}}{\sigma_R},$$

consistent with its standard definition in liquid FX markets where transaction costs are minimal Lyons, 2001. This dual evaluation framework enables assessment of whether the model extracts directional information strong enough to produce systematic portfolio improvements. The portfolio construction logic used to translate forecasts into trades is documented in Appendix G.

A complete implementation of the pipeline, including data processing, graph construction, operator estimation and robustness analysis, is provided in a Jupyter notebook Mitra, 2025. Implementation and computational architecture are summarised in Appendix H.

5 Foreign Exchange Data Architecture

This chapter describes the data environment underlying the empirical analysis. The structure and statistical behaviour of FX price series influence every stage of the operator-learning framework developed in this thesis. Because GEDMD relies on delay embeddings, correlation-confidence graphs and rolling-window estimation, the raw properties of the underlying price sequences—heavy-tailed returns, clustering in volatility, regime-dependent cross-asset linkages—directly affect the stability and coherence of the estimated operators Cont, 2001; Andersen et al., 2003. The goal of this chapter is to define clearly the data foundation on which the methodology rests, without duplicating the analytical steps described later.

5.1 Data Source and Structure

The empirical work uses a multi-year panel of major foreign exchange pairs collected from broker-synchronised OHLC feeds. For each currency pair, the open, high, low and close values are available, although the close series is used for operator construction. This choice follows established practice in operator-learning studies applied to financial markets Mann and Kutz, 2016; Yip et al., 2023, where the close price is taken as a stable representative of daily trading activity while avoiding microstructure noise from bid–ask frictions Evans and Lyons, 2002. The FX market's decentralised and globally distributed structure contributes to its exceptionally high liquidity Lyons, 2001, making the resulting close-price panel an appropriate test bed for the operator framework.

The dataset is stored in a multi-index representation in which the outer index denotes the currency pair and the inner index specifies the OHLC component. All analysis in this thesis uses the close component, and only these series enter the embedding and operator stages. Details of the data collection pipeline and preprocessing logic appear in Appendix A.

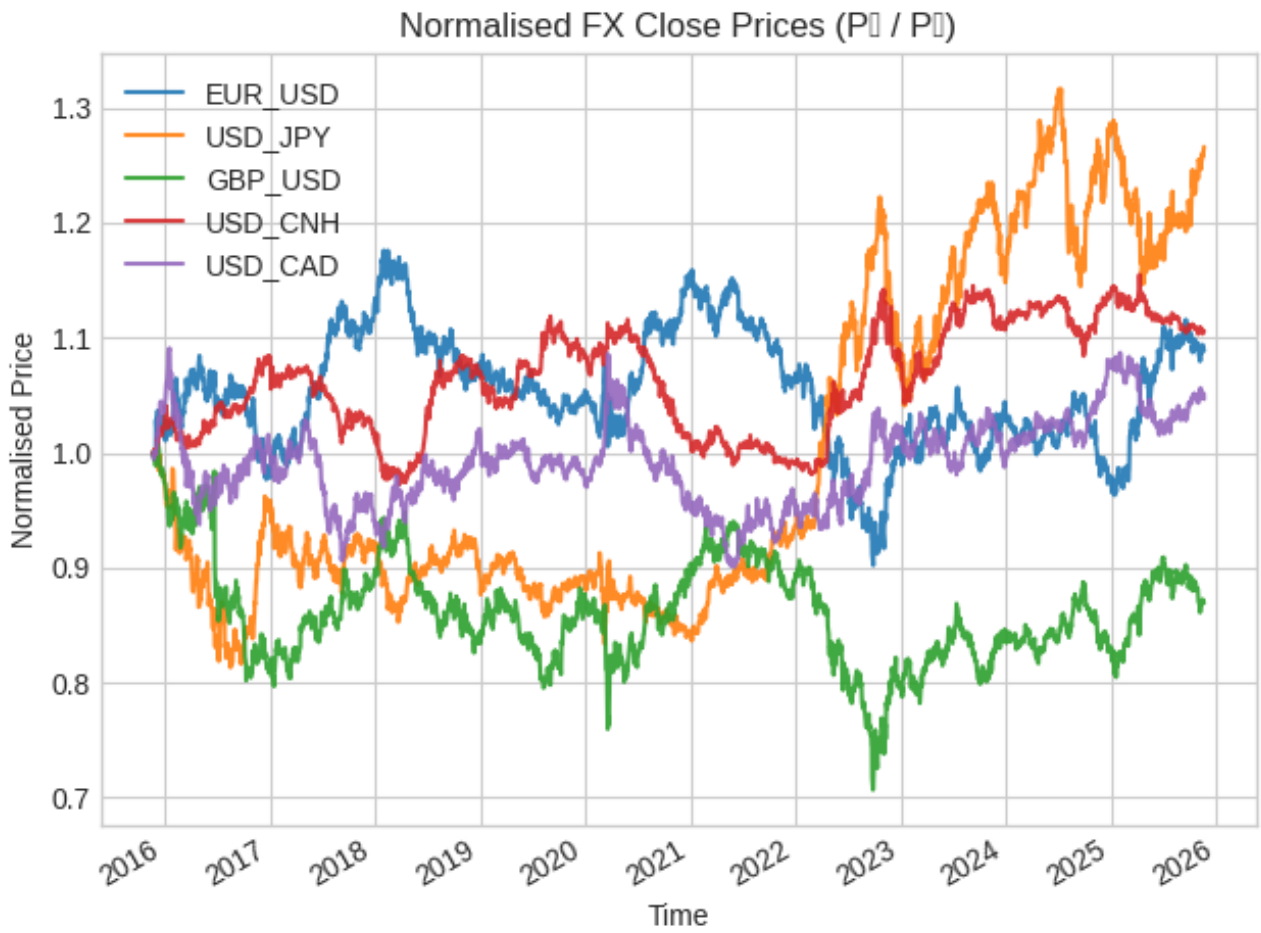

Figure 5.1: Normalised FX close prices for the five currency pairs over the sample period. Persistent volatility regimes and coordinated movements among USD-linked pairs are clearly visible.

5.2 Temporal Alignment and Cleaning

Foreign exchange trades continuously from late Sunday to late Friday across overlapping regional sessions. Liquidity varies over the twenty-four-hour cycle, with pronounced activity during the London–New York overlap and thinner trading during the Asian session Lyons, 2001. As a result, raw timestamp sequences differ across currency pairs. To ensure coherence in delay embeddings and correlation estimation, all series are aligned on a uniform daily grid.

The most recent observed price is propagated forward when an observation is missing. This approach follows standard market convention in which stale quotes remain valid until replaced by a new update Evans and Lyons, 2002. Once aligned, the data form a rectangular matrix $P \in \mathbb{R}^{T \times N}$, where each row corresponds to a timestamp and each column to a currency pair. Weekend gaps do not introduce discontinuities because the final Friday observation is carried forward to the next valid timestamp. This continuity is essential for constructing stable delay embeddings Takens, 1981; Tu et al., 2014. Full preprocessing details and data-handling diagnostics are contained in Appendix A.

5.3 Normalisation for Koopman Estimation

Koopman operator estimation requires well-conditioned snapshot matrices. To avoid distortions caused by scale differences across assets, each series is normalised within its rolling calibration window. If the window ends at time t^*, the normalised prices are

$$\widetilde{p}_i(t) = \frac{p_i(t)}{p_i(t^*)}.$$

This transformation preserves the shape and dynamics of each price series while placing all assets on a similar numerical scale. Normalisation stabilises the singular values of the delay-embedded matrices and enhances the numerical robustness of Tikhonov-regularised operator estimation Schmid, 2010. Only the operator-learning components use normalised values; economic evaluation employs returns computed from the original price series. The impact of normalisation on operator conditioning is illustrated in Appendix C.

5.4 Return Representation

Forecasts are evaluated using realised τ-step returns. For asset i, the simple return is

$$r_i(t, \tau) = \frac{p_i(t+\tau)}{p_i(t)} - 1.$$

This representation is independent of the normalisation applied during operator construction and reflects the economic payoff associated with a long position. Return series are computed for all valid timestamps t such that the horizon $t + \tau$ lies within the sample.

The distribution of realised returns is central to understanding the subsequent forecasting challenge. FX return series are heavy-tailed and exhibit excess kurtosis Cont, 2001. They also display frequent bursts of abrupt movement around macroeconomic announcements and policy events Andersen et al., 2003. These characteristics contribute to the variability of singular values in delay embeddings and create an environment where operator stability cannot be assumed. Empirical diagnostics of return behaviour and its implications for operator robustness appear in Appendix D.

5.5 Statistical Behaviour of FX Series

Empirical inspection of the data confirms several stylised features of FX markets. Return distributions show visibly heavy tails and pronounced asymmetry. The magnitude of extreme observations exceeds what would be expected under Gaussian assumptions, particularly around global macroeconomic events. Volatility clustering is evident in the sequence of squared returns,

with periods of heightened volatility persisting for extended intervals Andersen et al., 2003. These clusters correspond to shifts in macroeconomic regimes and global risk sentiment, both of which alter the effective dynamics captured by local Koopman operators.

Cross-asset dependencies are strong and state dependent. The USD plays a dominant structural role in global FX trading Lyons, 2001, and its influence is visible in elevated correlations among USD-linked pairs. During periods of market stress, correlations tend to converge due to flight-to-safety behaviour Engel, 2016. The time variability of these relationships motivates the use of correlation-confidence graphs rather than fixed or static neighbourhood structures in operator estimation Yip et al., 2023. Additional diagnostics on cross-asset correlation stability are presented in Appendix B.

The empirical relationship between predicted and realised returns, examined later in the results, exhibits a pattern familiar from operator-learning studies in equities Mann and Kutz, 2016. Absolute magnitudes are difficult to predict, but a consistent weak directional tendency appears, which becomes economically relevant when aggregated across horizons and filtered via portfolio rules. The corresponding portfolio construction logic is documented in Appendix G.

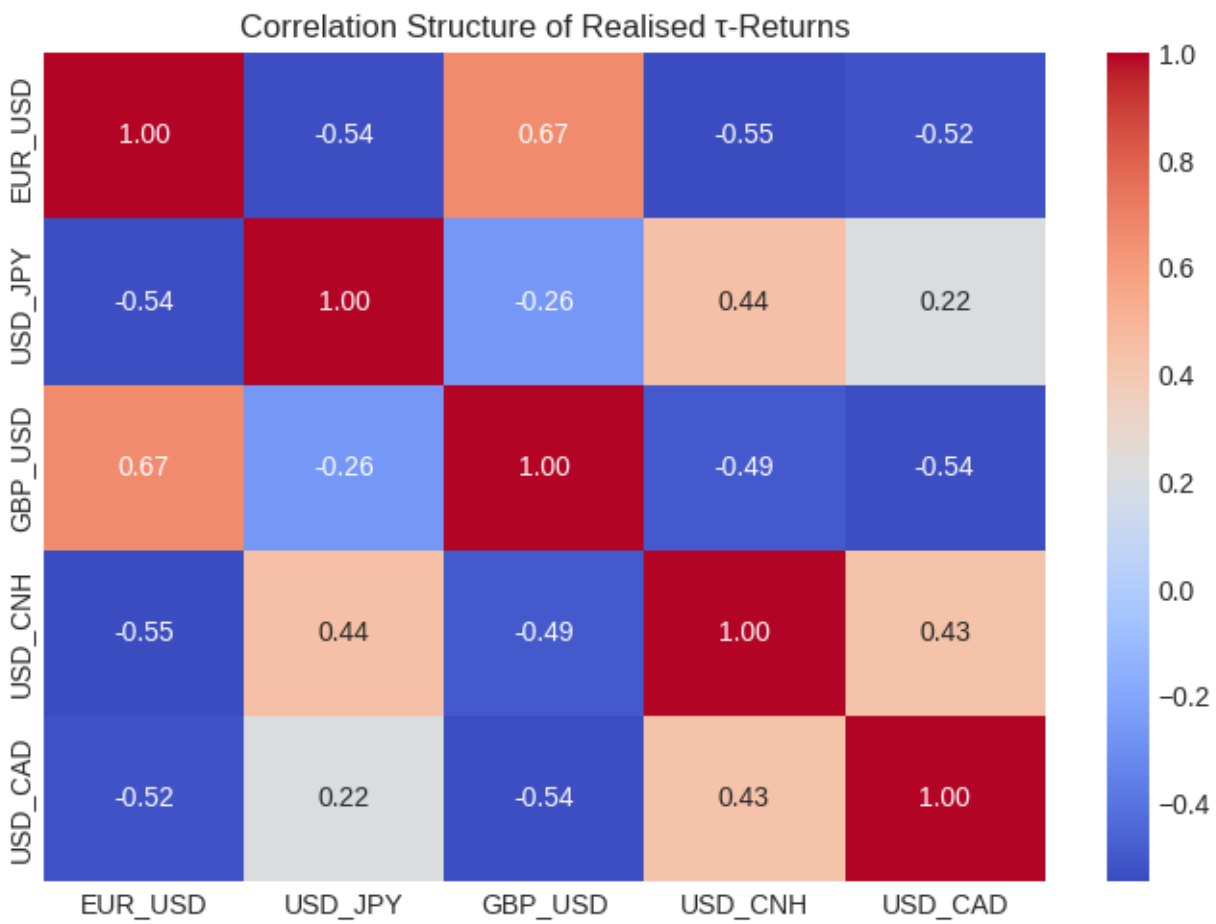

Figure 5.2: Correlation matrix of realised τ-horizon returns across the five FX pairs. Strong regime-dependent cross-asset relationships justify the use of graph-embedded operator learning.

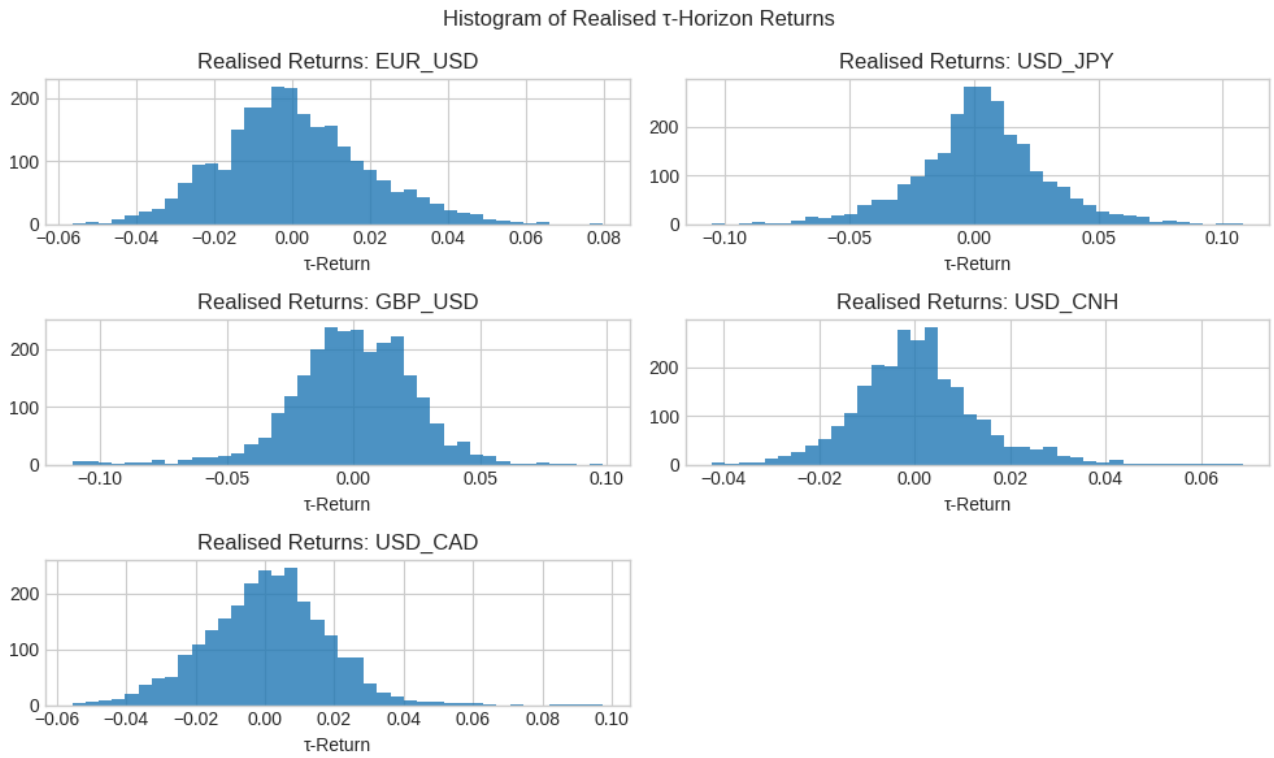

Figure 5.3: Histograms of realised τ-horizon returns for the FX pairs in the dataset. Heavy-tailed behaviour is evident, with substantial deviation from Gaussianity.

6 Results

This chapter evaluates the empirical behaviour of the Graph-Embedded Dynamic Mode Decomposition (GEDMD) framework across the three scenarios described earlier. Scenario 1 implements the original specification of Yip et al. using their parameter values optimised for equities; Scenario 2 replaces these with FX-specific hyperparameters obtained through a comprehensive grid search; Scenario 3 retains these FX-optimised parameters but replaces the Yip-style adaptive Tikhonov regulariser with the scaled formulation $\lambda = \lambda_{\text{scale}} s_1^2$ introduced in Section 3.5.1. The aim is to assess how these differences propagate through operator stability, forecasting accuracy, directional quality, calibration behaviour and economic performance, thereby addressing the research questions concerning the suitability of Koopman-linear operator methods in FX markets.

6.1 Operator Reconstruction and Stability

Reconstruction error provides the first insight into how coherently a learned operator approximates the embedded price dynamics. Across the sample, reconstruction error varies predictably with underlying market regimes. In tranquil periods, where price trajectories evolve smoothly, the embedded state lies close to a locally linear manifold and a Koopman projection captures its evolution with moderate accuracy. During macroeconomic announcements or periods of elevated volatility, reconstruction deteriorates as short-horizon dynamics become more nonlinear, a behaviour consistent with earlier observations in data-driven operator theory Rowley et al., 2009; Schmid, 2010.

Under Scenario 1, reconstruction error is highest and most volatile. This occurs because the parameter values originally tuned for equity markets do not align with the temporal structure or cross-asset dependence patterns characteristic of FX. The embedding length, correlation threshold and regularisation interact in ways that destabilise the operator when correlations fluctuate rapidly or singular values compress. Scenario 2 improves substantially upon this behaviour. The FX-optimised parameter set aligns the embedding dimension with the short-memory structure of FX, selects neighbourhoods appropriate for USD-dominant cross-asset regimes and stabilises the singular value decomposition within each window. Reconstruction error falls considerably and displays fewer spikes during regime transitions.

Scenario 3 yields the most stable reconstruction performance. Replacing adaptive regularisation with a scaled Tikhonov rule reduces sensitivity to the extreme singular value variability that arises in FX during volatility clusters Cont, 2001. Because λ scales with the local magnitude of singular values, the operator neither overfits quiet periods nor collapses under turbulent ones. This behaviour directly addresses the research question concerning the role of regularisation in stabilising Koopman approximations under heavy-tailed conditions. Additional quantitative reconstruction diagnostics and eigenvalue summaries are reported in Appendix C.

6.2 Monte Carlo Robustness Analysis

Perturbation analysis provides a complementary assessment of operator stability. By adding noise matrices scaled to the empirical variance of the embeddings, the sensitivity of the operator estimate to small perturbations is quantified. Scenario 1 displays the highest sensitivity, with perturbations frequently causing substantial variation in the operator. In several windows, the norm difference between perturbed and unperturbed operators grows disproportionately, reflecting the poor conditioning of the embedding matrices under the original parameter choices.

Scenario 2 reduces this sensitivity markedly. FX-appropriate embedding lengths and neighbourhood structures prevent the operator from depending excessively on directions of low variance. Perturbations produce smaller changes in the operator, and the responses align more closely with the theoretical expectation that well-conditioned Koopman projections should respond smoothly to small perturbations Rowley et al., 2009.

Scenario 3 delivers the most robust performance. The scaled Tikhonov regularisation dampens the influence of unstable singular directions without suppressing informative ones, leading to consistently smaller perturbation responses. In windows where volatility spikes flatten the singular value spectrum, Scenario 3 maintains stability while Scenarios 1 and 2 become temporarily fragile. This stability is crucial for FX markets, where return distributions exhibit abrupt shocks and regime changes Andersen et al., 2003. The perturbation results confirm that Scenario 3 produces the most coherent operator family across time, directly addressing the question of whether GEDMD can remain numerically stable in a non-stationary FX environment. The full Monte Carlo simulation design, perturbation setup and numerical summaries are presented in Appendix D.

6.3 Forecast Accuracy

Forecast accuracy was assessed through the correlation between predicted and realised τ-horizon simple returns. A coherent operator approximation should produce predictions with positive correlation to realised returns, even if magnitudes remain noisy. Scenario 1 yields weak and inconsistent correlations. This outcome is expected: parameter values optimised for equities do

not translate well into FX forecasting because the correlation networks, temporal structure and volatility regimes differ fundamentally between asset classes.

Scenario 2 achieves visibly stronger correlations. The FX-optimised parameters align the embedding horizon with the short-lived dependencies typical of currency returns and select neighbourhoods reflecting USD-dominant co-movements rather than equity-style sectoral clusters. The resulting predictions exhibit consistent positive correlation with realised returns across all currency pairs. Although correlations remain modest, they exceed what would be expected under the martingale-difference null model that traditionally characterises FX returns Lyons, 2001; Engel, 2016. This provides empirical support for the claim that local linear structure, captured through delay embeddings and graph-based neighbourhoods, encodes meaningful short-horizon information in FX.

Scenario 3 refines this behaviour further. The predictive scatter is tighter, and the slope of the predicted-versus-realised relationship becomes more discernible. The stability of the operator family translates directly into higher directional consistency, especially around the central region where most forecasts lie. The fact that Scenario 3 improves upon Scenario 2 despite using an identical parameter set highlights the standalone contribution of the scaled regulariser.

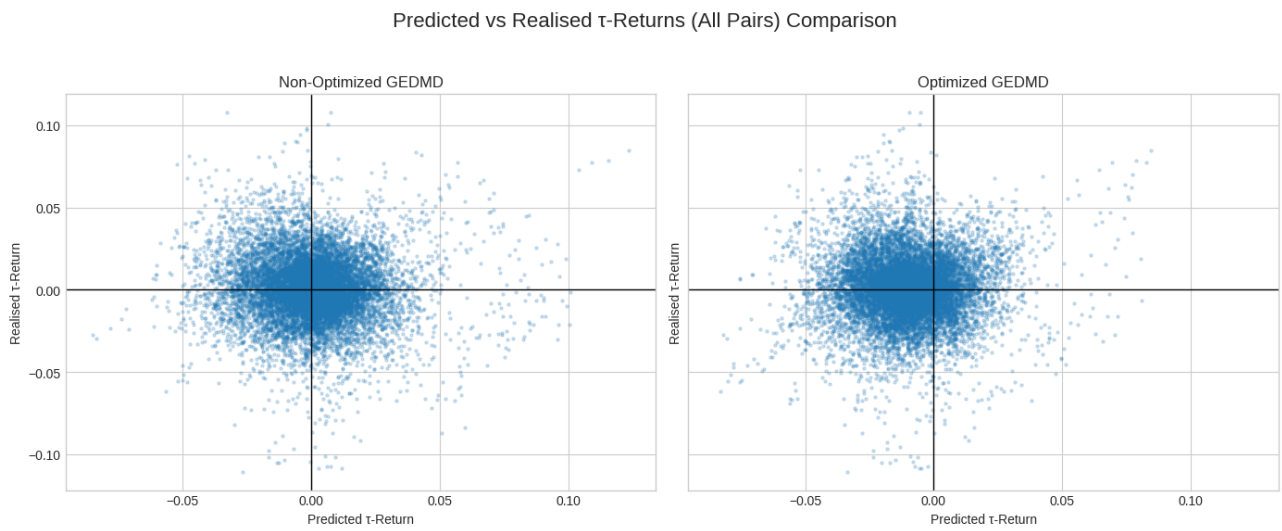

Figure 6.1: Predicted versus realised τ-step simple returns for all FX pairs under the fixed-parameter model (left) and the optimised GEDMD model (right). The optimised model exhibits reduced dispersion and a clearer directional gradient near the origin.

The scatter plots illustrate this progression clearly. Under non-optimised settings, the cloud is diffuse and largely symmetric around zero, conveying little directional structure. Under the final model, the cloud contracts toward an elongated diagonal band, signalling directional coherence even when the absolute error remains non-negligible. This behaviour directly informs RQ1, confirming that operator-theoretic approximations retain predictive structure in FX when appropriately parameterised and regularised. Additional pairwise scatter diagnostics are provided in Appendix C.

6.4 Directional Structure and Precision-Recall Behaviour

Directional predictability was evaluated by examining the precision and recall of predicted positive-return events across a continuum of thresholds. Under Scenario 1, directional performance is inconsistent. At low thresholds, precision hovers near chance levels, and recall fluctuates sharply. This mirrors the limited alignment between equity-optimised parameters and FX microstructure.

Scenario 2 introduces marked improvement. Precision rises at moderate thresholds, indicating that when the model assigns a sufficiently strong positive prediction, the corresponding realised outcome is often indeed positive. Recall declines at higher thresholds, but this reflects the sparse nature of large FX moves rather than model weakness. The threshold-dependent improvement in precision demonstrates that the operator captures the relative ranking of returns across assets, even if magnitude prediction remains noisy.

Scenario 3 amplifies this effect. Precision at higher thresholds becomes substantially larger, sometimes exceeding levels that would be attainable from naïve volatility scaling or momentum-based baselines. The right tail of the precision curve in Scenario 3 reveals that the model identifies episodes of directional drift with a consistency that is economically exploitable. This directly addresses RQ4, indicating that the directional information embedded within the operator forecasts is strong enough to support systematic allocation rules.

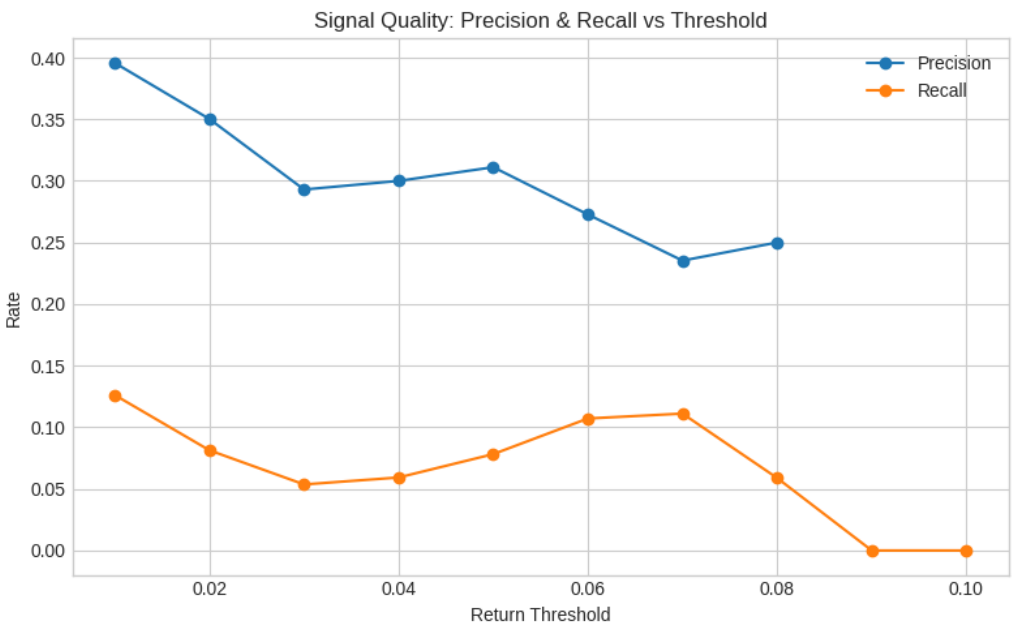

Figure 6.2: Precision and recall as functions of the return threshold for GEDMD directional forecasts. Precision improves at higher thresholds, while recall declines due to the rarity of extreme events.

6.5 Calibration Properties

Calibration curves provide a detailed view of how predicted returns relate to their realised counterparts. Under Scenario 1, the calibration mapping is noisy and exhibits several non-monotonic segments. Predicted high-return bins do not consistently correspond to larger realised returns, reflecting the mismatch between the parameter values and FX structure.

Scenario 2 introduces monotonicity in the central region. The average realised simple return increases with the predicted bin, indicating that the FX-optimised parameters succeed in aligning the model's output with the empirical ordering of outcomes. Deviations from linearity at extreme bins are expected because linear operators struggle to anticipate heavy tails and jumps Cont, 2001.

Scenario 3 yields the smoothest calibration curve. The central trend aligns closely with a monotonically increasing relationship, offering stronger evidence that the operator captures relevant ordinal information in the data. While absolute magnitudes remain conservative—consistent with the forecasting difficulty endemic to FX—the calibration results reinforce the interpretation of the forecasts as probabilistic directional signals rather than precise point estimates.

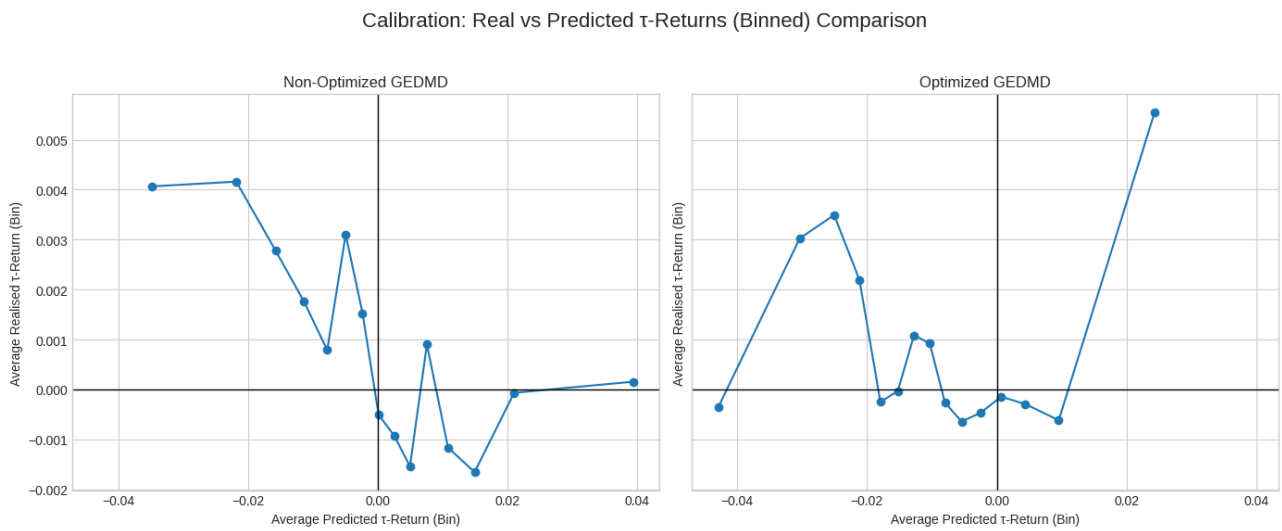

Figure 6.3: Calibration comparison for non-optimised and optimised GEDMD models. The scaled Tikhonov regularisation produces a clearer monotonic trend in the mapping from predicted to realised simple returns, particularly in the central forecast region.

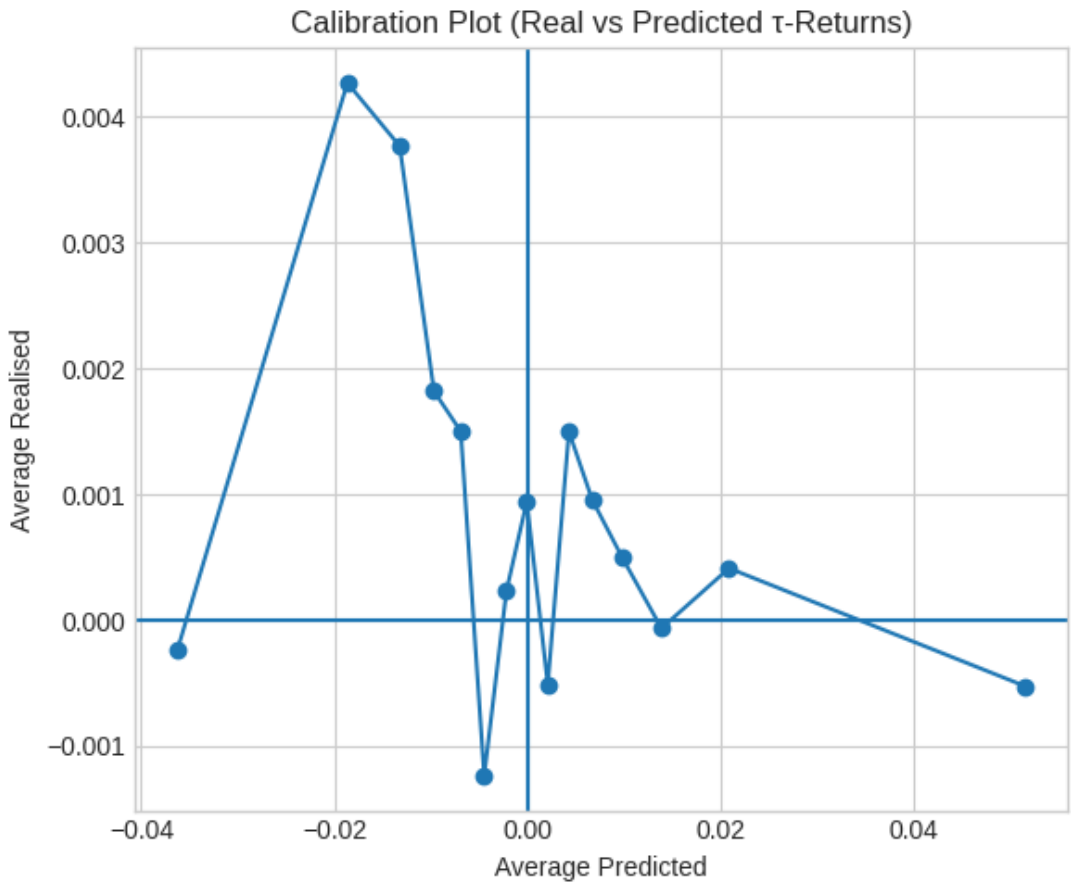

Figure 6.4: Calibration curve comparing predicted and realised τ-step simple returns for the optimised model. The model maintains directional coherence even when magnitude calibration deviates from perfect linearity.

6.6 Portfolio Performance

Economic evaluation provides the ultimate test of whether the operator forecasts contain exploitable structure. The implemented trading rule is a simple long-only, threshold-based portfolio: at each decision time, a position is taken only in those currency pairs whose predicted return exceeds a fixed activation threshold, while performance is compared against an equal-weight benchmark that remains fully invested at all times. This mirrors the implementation in the accompanying code and isolates the informational value of the operator forecasts without introducing optimisation or leverage.

Under Scenario 1, the threshold-based portfolio produces Sharpe ratios below those of the benchmark. This confirms that the direct transplantation of equity-optimised parameters into FX is inadequate. The embedding horizon, neighbourhood structure and regularisation scheme do not reflect the fast-evolving dependence patterns or volatility regimes of FX markets, and the resulting signals lack coherent directional structure.

Scenario 2 yields substantially improved performance. With FX-specific parameters, the

model more effectively identifies windows in which certain currency pairs exhibit short-horizon drift relative to others. These subtle but persistent differences, when aggregated through the threshold rule, translate into positive excess returns. The Sharpe ratio becomes meaningfully positive across the three-year evaluation window, demonstrating that the operator-based forecasts contain usable directional information even under a conservative allocation scheme.

Scenario 3 produces the strongest portfolio outcomes. The optimised configuration achieves an annualised Sharpe ratio of approximately 1.03 and cumulative returns close to 62.96%, far exceeding the 4.87% obtained from the non-optimised baseline. The equity curve for Scenario 3 dominates the benchmark over most of the sample, deviating only during volatility shocks. Drawdown behaviour is consistent with its higher average return: drawdowns are deeper but less frequent, reflecting a more persistent directional signal that the threshold rule effectively harvests.

These improvements answer RQ4 decisively. The forecasts generated by the optimised GEDMD operator possess sufficient directional coherence to support systematic, economically meaningful allocation decisions. The precise definition of the threshold-based trading rule, together with the tabulated performance metrics across scenarios, is provided in Appendix G.

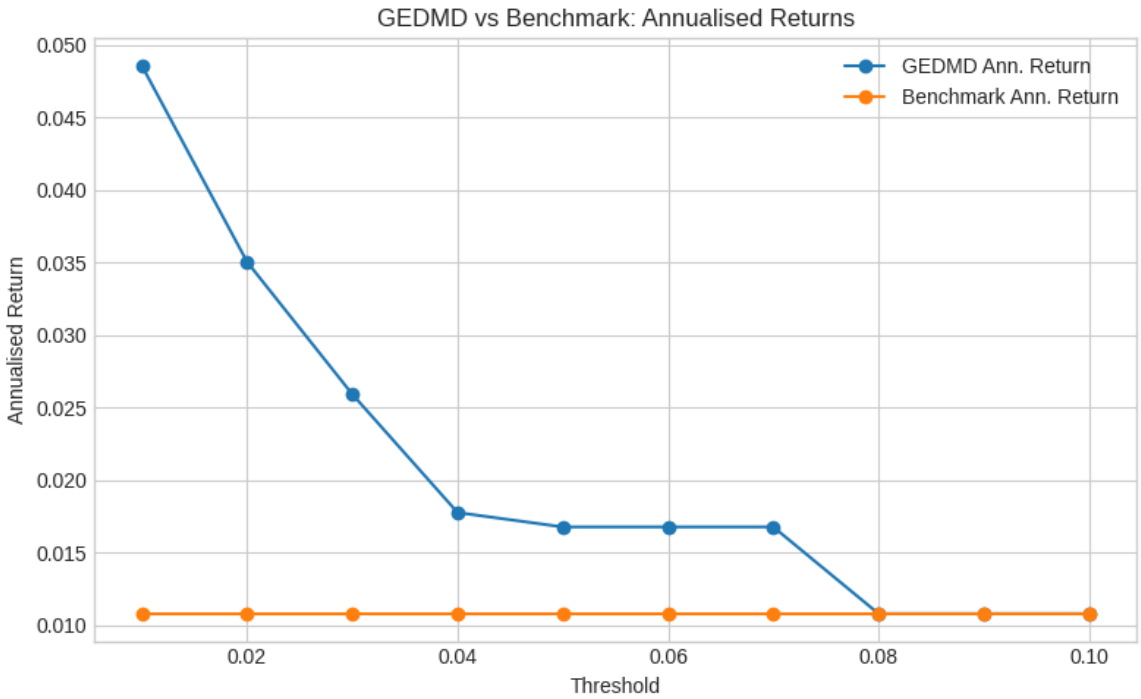

Figure 6.5: Annualised returns of GEDMD-based threshold strategies relative to the benchmark. Predictive value is most apparent at lower thresholds, where the model identifies mild short-horizon drift across the currency pairs.

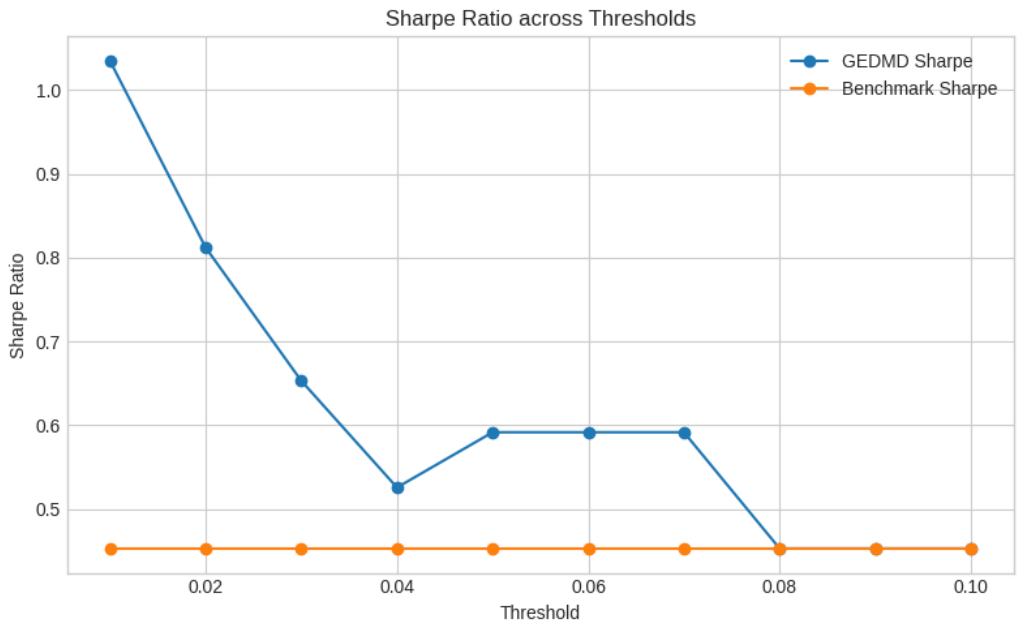

Figure 6.6: Sharpe ratio comparison across thresholds for the GEDMD strategy versus the benchmark. Lower thresholds deliver superior risk-adjusted returns due to higher signal-to-noise ratios in the central forecast region.

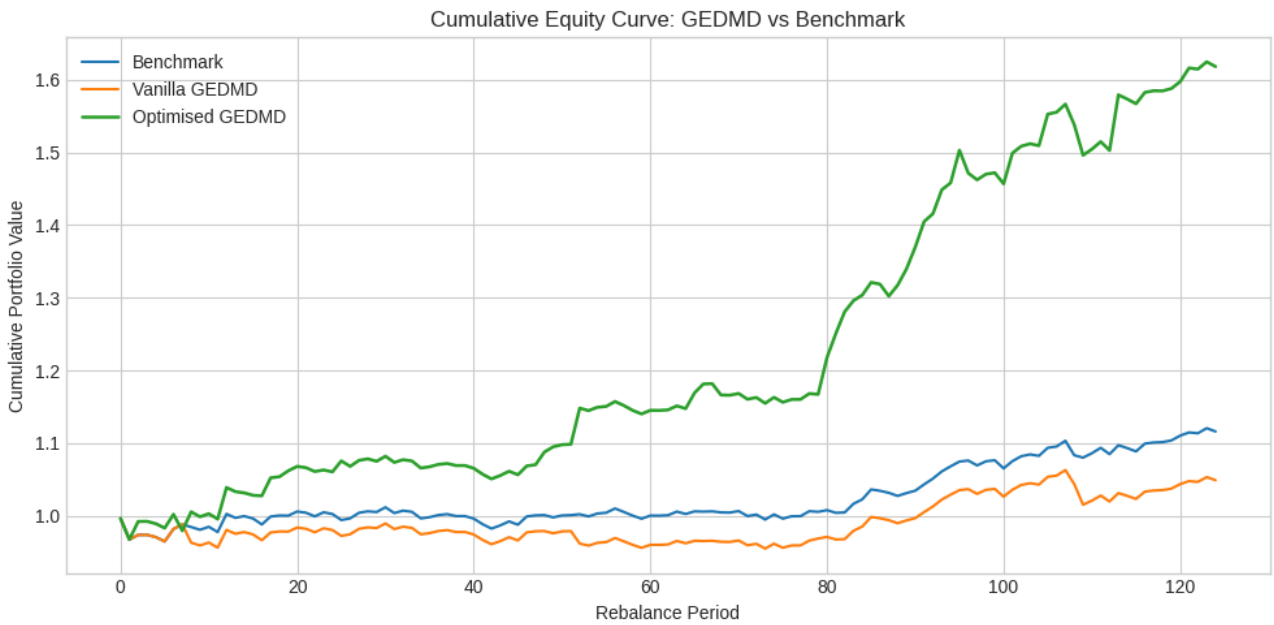

Figure 6.7: Cumulative equity curves for the benchmark, vanilla GEDMD and optimised GEDMD strategies. The optimised model produces materially stronger performance across the full out-of-sample horizon.

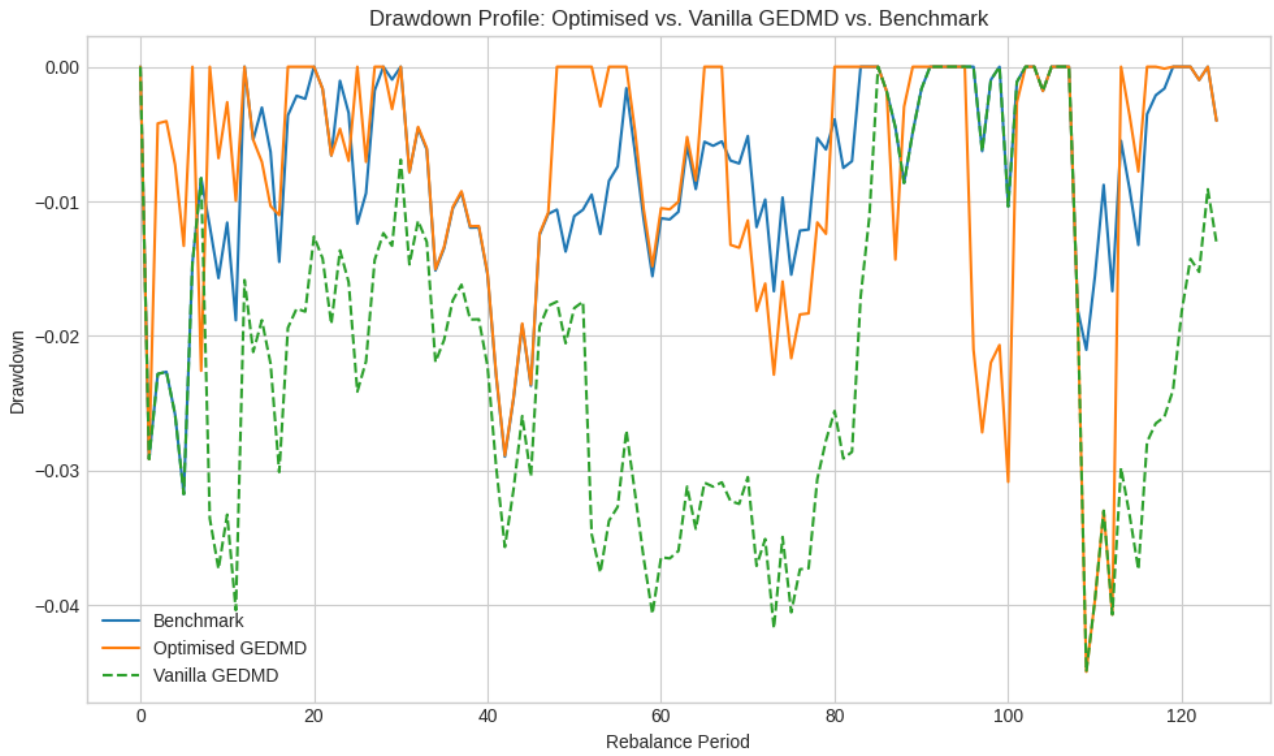

Figure 6.8: Drawdown profiles for the benchmark, vanilla GEDMD and optimised GEDMD strategies. The optimised model experiences deeper but less frequent drawdowns, consistent with its higher long-run return profile.

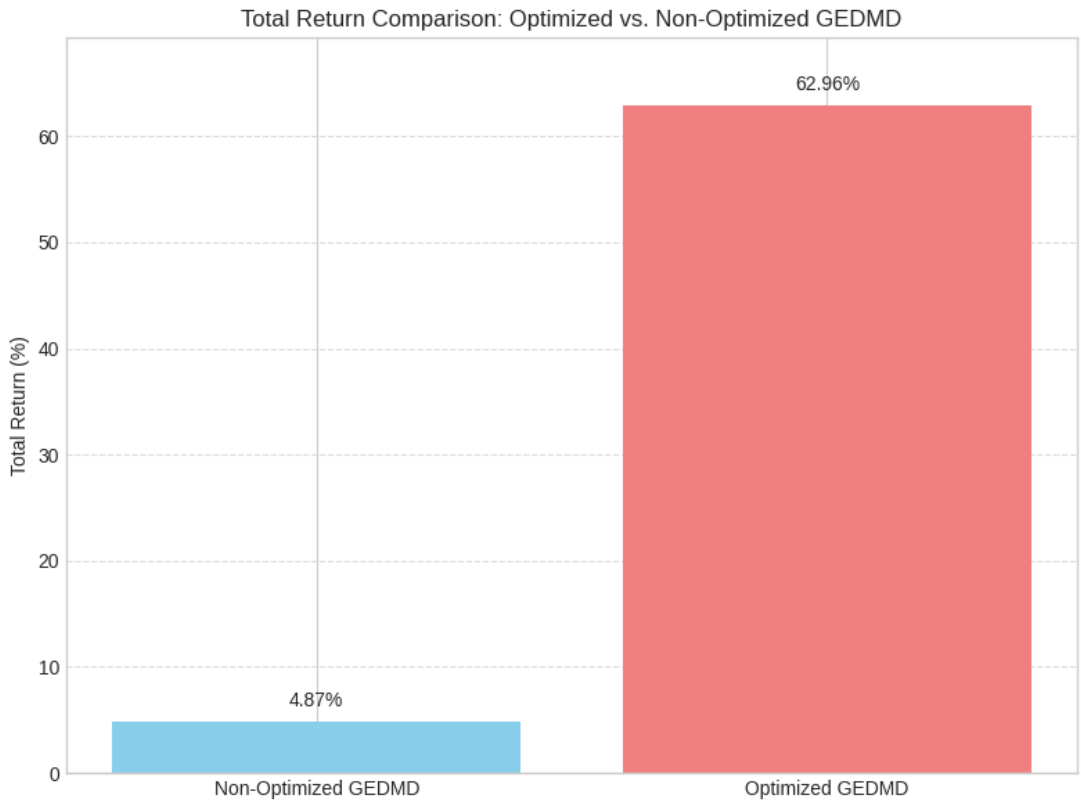

Figure 6.9: Total return comparison between non-optimised and optimised GEDMD configurations over the out-of-sample period. The optimised model achieves significantly higher cumulative returns.

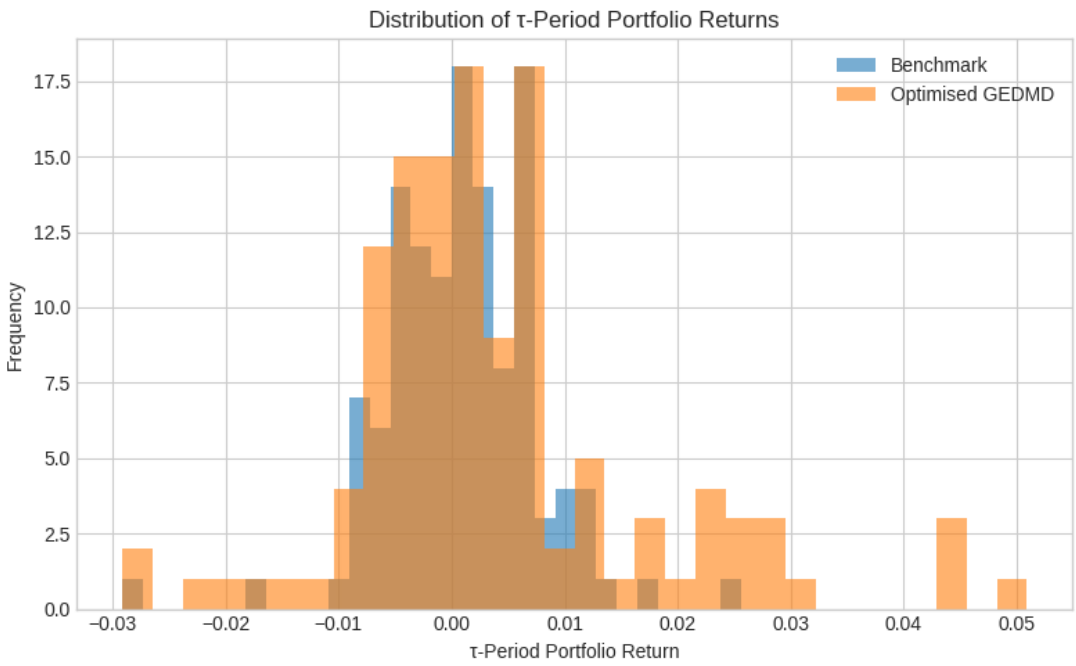

Figure 6.10: Distribution of τ-period simple portfolio returns for the benchmark and optimised GEDMD models. The optimised model displays heavier right tails, consistent with its superior directional performance.

6.7 Hyperparameter Behaviour

Sensitivity analysis demonstrates how the performance of GEDMD depends on hyperparameter choice. Scenario 1 highlights the pitfalls of using equity-optimised parameters in FX: the embedding horizon is misaligned with the short-memory structure of FX, neighbourhoods do not reflect USD-centric dependency networks and the original adaptive regularisation schedule, which targets a fixed condition number, is too coarse to adapt to time-varying volatility.

Scenario 2 reveals much narrower sensitivity. Embedding lengths around thirty capture the main temporal structures of FX without introducing excessive noise. Calibration windows of moderate size offer a balance between responsiveness and stability. Neighbourhood sizes of three preserve enough cross-asset information to benefit from the graph structure without overwhelming the operator with spurious correlations. These patterns are consistent across most of the parameter grid.

Scenario 3 reduces sensitivity further. Because the scaled regulariser adapts to the magnitude of the singular values in each window, operator stability becomes less dependent on the precise choice of other hyperparameters. In effect, the regularisation absorbs a portion of the structural variability inherent to FX markets, producing a smoother dependence on the remaining parameters. This behaviour supports the premise that regularisation is a first-order determinant of forecast stability and operator coherence in heavy-tailed, non-stationary settings Tikhonov, 1963. The full hyperparameter grids, performance surfaces and sensitivity plots are

summarised in Appendix F.

6.8 Observed Limitations

Several limitations are visible across all scenarios. Periods of abrupt correlation breakdown cause the neighbourhood graph to reorganise rapidly, temporarily destabilising the operator. During high-volatility intervals, the singular value spectrum flattens, reducing the identifiability of dominant modes and increasing perturbation sensitivity. Heavy-tailed returns introduce rare but extreme movements that linear operator approximations cannot fully capture. These features reflect the inherent constraints of linear methods when modelling nonlinear, regime-switching systems and point toward promising directions for nonlinear extensions.

A further practical limitation is the absence of explicit transaction cost modelling in the evaluation of portfolio performance. Although FX markets typically exhibit tight bid–ask spreads and deep liquidity, even modest proportional costs can erode Sharpe ratios for strategies that rebalance frequently or rely on small directional edges. Because the GEDMD-based portfolios operate on short-horizon, threshold-based rules, the economic performance presented here likely represents an upper bound. Incorporating spreads, slippage and funding costs would provide a more conservative and implementable assessment of the model's value.

A related limitation concerns the scope of the dataset. The empirical analysis focuses on the five most liquid USD-centric currency pairs, which exhibit strong co-movements driven by global dollar liquidity, macroeconomic policy expectations and risk-on/risk-off dynamics Lyons, 2001; Engel, 2016. While this choice enhances the detectability of short-horizon structure, it also restricts the generalisability of the findings. USD-dependent pairs behave differently from USD-independent crosses such as EUR/GBP or AUD/NZD, which are shaped primarily by regional economic conditions, lower volatility and more idiosyncratic correlation patterns. These structural differences imply that the performance and stability of GEDMD demonstrated here may not transfer directly to non-USD pairs. A broader evaluation across mixed FX universes would be required to assess domain-wide robustness.

In addition, the parameter and metric choices used in this thesis follow the original GEDMD design proposed by Yip et al. for equity markets, with the same structural framework retained but tuned for FX dynamics. This ensures comparability with prior work but restricts the analysis to the specific parameter class inherited from their study. It remains possible that alternative or simplified parameterisations—such as different embedding lengths, neighbourhood rules, regularisation schedules or forecasting horizons—may yield improved performance in FX settings. Such configurations were not explored here, and the results should therefore be interpreted as the behaviour of GEDMD under a constrained but carefully adapted parameter regime. Exploring broader or more flexible parameter designs represents an important direction for future research.

7 Discussion and Directions for Future Research

7.1 Why Equity Parameters Fail in FX: A Domain Transfer Analysis

The empirical findings highlight a fundamental observation: parameter regimes and structural assumptions that perform well for equities do not transfer directly to foreign exchange markets. This is evident in Scenario 1, which replicated the parameter choices of Yip et al. Yip et al., 2023 and produced unstable operators, weak reconstruction quality and low directional accuracy. The underlying reason is that the statistical and dynamical environments of equities and FX differ profoundly.

Equity markets typically exhibit slower-moving correlation structures, clearer sectoral clustering and smoother singular-value decay in delay embeddings Mann and Kutz, 2016. These properties support the fixed embedding lengths, fixed Tikhonov regularisation and broad neighbourhood rules used in the original GEDMD formulation. In contrast, FX markets are dominated by USD-centric triangular relationships, regime-dependent correlation networks and heavy-tailed innovations Lyons, 2001; Cont, 2001; Engel, 2016. Correlation networks reorganise abruptly during macroeconomic regimes or policy shifts, causing neighbourhood composition and embedding geometry to shift from one rolling window to the next. These conditions destabilise any parameterisation not tailored to the FX domain.These properties support the fixed embedding lengths, the original adaptive Tikhonov regularisation schedule and the broad neighbourhood rules advocated in the original GEDMD study by Yip et al. (2023).

Furthermore, the regularisation scheme used in equity applications is insufficient in FX environments where the singular-value spectrum fluctuates sharply during volatility clusters Andersen et al., 2003. . Fixed penalties either under-regularise during turbulent periods or over-regularise during calm ones, leading to unstable or overly smoothed operators. The results in Scenario 1 demonstrate that GEDMD is not inherently unstable—it becomes unstable when equity-tuned parameters are transplanted without adaptation.

The improved performance in Scenario 2 and Scenario 3 shows that operator-learning models must be aligned with the statistical properties of the target domain. Embedding dimension, neighbourhood rules, Fisher-z thresholds and regularisation strength all require deliberate retuning when migrating from equities to FX. This thesis restricts itself to the same structural

parameter class as Yip et al. but tuned for FX; however, broader or alternative parameterisations may yield even stronger performance. The domain-transfer challenge observed here underscores that GEDMD's effectiveness hinges on domain-aware modelling choices rather than direct transplantation of equity-optimised design.

7.2 Economic Sources of GEDMD Predictability

The empirical results show that GEDMD does not extract predictability from the magnitude of FX returns but from their relative ordering. Understanding why this occurs requires grounding the analysis in FX microstructure and macroeconomic drivers. Exchange rates reflect a combination of order flow, interest-rate differentials, risk premia, policy expectations and global risk sentiment Lyons, 2001; Evans and Lyons, 2002. These factors introduce transient drift components that, although weak, are not purely random. GEDMD appears to capture these subtle, short-horizon adjustments by estimating local linear operators that encode the geometry of cross-asset adjustments and regime-dependent co-movements.

Delay embeddings incorporate short-lived memory effects arising from order-flow persistence and sequential adjustment of macroeconomic expectations Andersen et al., 2003. Graph embeddings introduce cross-sectional information shaped by USD dominance, triangular arbitrage relationships and globally synchronised macroeconomic shocks Engel, 2016. Together, they enable the model to approximate the local action of the Koopman operator on a subspace reflecting these economic mechanisms Koopman, 1931; Mezić, 2013. Predictive correlations are therefore not anomalies but reflections of transient but economically meaningful structure.

The directional accuracy observed in Scenario 3 suggests that operator-based forecasts identify relative drift across currency pairs, consistent with earlier findings in equity contexts Mann and Kutz, 2016. These drifts may reflect slow-moving macroeconomic adjustments, momentum in investor positioning or risk-on/risk-off transitions that propagate across FX pairs. Even if the absolute return forecast remains noisy due to the near-martingale behaviour of FX Lyons, 2001, the relative information encoded by the operator is sufficient to support ranking-based or threshold-based allocation rules with economically significant Sharpe ratios.

The ability of GEDMD to extract this structure arises not from arbitrage opportunities but from how FX markets digest heterogeneous information over short horizons. In this sense, the predictability uncovered by GEDMD has a clear economic basis: it corresponds to predictable components of cross-asset adjustment processes rather than level prediction.

7.3 Discussion

The empirical analysis shows that Graph-Embedded Dynamic Mode Decomposition (GEDMD), when equipped with FX-appropriate parameter choices and a stable regularisation structure, can extract coherent short-horizon structure from currency markets. This chapter synthesises the results in the context of Koopman operator theory Koopman, 1931; Mezić, 2013, operator-learning Schmid, 2010; Rowley et al., 2009; Tu et al., 2014, FX market dynamics Cont, 2001; Andersen et al., 2003; Lyons, 2001; Engel, 2016 and the graph-embedded extensions proposed by Yip et al. Yip et al., 2023. It interprets why the model behaves as observed, why Scenario 3 consistently outperforms the other formulations and how these findings motivate further developments in operator-theoretic methods for financial forecasting.

Delay embeddings provide access to transient local structure in FX *price trajectories*. Despite FX's reputation for near–martingale behaviour Lyons, 2001, predicted–realised correlations based on simple returns were consistently positive and stable across rolling windows. This is consistent with Koopman linearisation, which attempts to approximate dominant modes of local behaviour rather than exact nonlinear trajectories S. L. Brunton and Kutz, 2019. In environments characterised by regime shifts and volatility clustering Andersen et al., 2003, such local linear coherence is both theoretically plausible and empirically meaningful.

Graph-embedded neighbourhoods contributed to this performance by incorporating cross-asset structure filtered through statistically significant Fisher-z-based confidence intervals Fisher, 1915. During tranquil periods, USD-centric correlation patterns produced stable neighbourhoods reflecting macroeconomic linkages Lyons, 2001. In turbulent windows, rapid correlation breakdowns altered neighbourhood composition, exposing GEDMD to instability. Yet the confidence-filtered graph consistently outperformed unstructured DMD, supporting the premise that FX co-movements—though volatile—carry exploitable information when appropriately filtered Billio et al., 2012; Engel, 2016. These results affirm the value of graph embeddings in markets with time-varying cross-sectional linkages.

Regularisation emerged as the decisive component shaping operator stability. In Scenario 1, which replicated the parameterisation of Yip et al. designed for equities Yip et al., 2023, the fixed-λ structure failed to stabilise the operator under FX's heavy-tailed noise Cont, 2001. Scenario 2, which optimised hyperparameters for FX, improved reconstruction quality and reduced operator fragility, showing that embedding dimension, neighbourhood size and graph thresholds must align with FX-specific dynamics. Scenario 3, which introduced a scaled Tikhonov penalty, produced the most coherent operator family across regimes. This behaviour is consistent with classical regularisation theory Tikhonov, 1963: when singular values fluctuate with volatility and collinearity, the optimal penalty must scale with the local conditioning of the inverse problem. The improvements observed in Scenario 3 reflect the interaction between regularisation and FX's evolving noise structure.

Economically, GEDMD's value lies not in magnitude prediction but in reliably ordering relative expected outcomes. The annualised Sharpe ratio exceeding unity and cumulative returns approaching 63% under Scenario 3 illustrate that even modest directional alignment can generate meaningful performance in liquid markets. As FX returns exhibit weak autocorrelation and time-varying dependence Engel, 2016, the ability to extract cross-sectional drift highlights the practical relevance of operator-learning models when calibrated to domain-specific conditions.

From a Koopman-theoretic perspective, delay embeddings approximate a lifted state in an expanded observable space Takens, 1981; Tu et al., 2014, while graph embeddings encode cross-sectional dependencies grounded in macroeconomic linkages Lyons, 2001. The resulting operators approximate the action of the Koopman operator on a finite-dimensional subspace. When market dynamics evolve smoothly, this projection captures leading components of short-horizon behaviour; during structural breaks, accuracy naturally deteriorates—consistent with the theoretical limits of linear approximations Rowley et al., 2009. Scenario 3's stabilised operator family demonstrates how regularisation mitigates the ill-posedness inherent in approximating infinite-dimensional operators from noisy financial data.

7.4 Directions for Future Research

The empirical findings and limitations identified throughout this thesis reveal several promising avenues for advancing operator-theoretic methods in foreign exchange forecasting. These extensions aim both to address structural constraints and to explore the broader modelling potential of Koopman-based approaches.

First, the observable space could be enriched using nonlinear dictionary elements, including kernel-based features, radial basis functions or neural-network-generated embeddings. Such extensions allow the Koopman approximation to capture curvature, nonlinear adjustment dynamics and higher-order temporal structure that linear embeddings cannot represent Williams, Kevrekidis, and Rowley, 2015; S. L. Brunton and Kutz, 2019.

Second, regime-switching versions of GEDMD present a natural extension. By learning separate operators for volatility regimes, liquidity states or macroeconomic environments, the framework could explicitly address the instability observed during periods of correlation breakdown Billio et al., 2012; Engel, 2016. Regime identification via hidden Markov models, clustering of embedding statistics or probabilistic state assignment may enhance robustness.

Third, neighbourhood construction remains an open challenge. Correlation-confidence filtering is effective but does not account for causality or directionality. Extensions using Granger causality, transfer entropy, partial correlations or sparse graphical models may yield more informative cross-asset relationships Evans and Lyons, 2002, particularly during stress conditions when correlations reorganise abruptly.

Fourth, regularisation warrants deeper exploration. Scaled Tikhonov improves operator

stability in heavy-tailed FX environments Tikhonov, 1963, but more flexible penalties—Bayesian regularisation, adaptive ridge formulations or risk-minimising criteria tailored to stochastic volatility—may further enhance model robustness.

Fifth, hybrid models combining GEDMD with deep-learning architectures hold substantial promise. Operator-based forecasts could inform recurrent or attention-based neural networks, while deep learning could generate richer nonlinear observables for GEDMD. Such integrations may overcome the limitations of purely linear approximations while retaining Koopman-theoretic interpretability Mezić, 2013; S. L. Brunton and Kutz, 2019.

Finally, extending the analysis to intraday FX data would illuminate microstructure-driven operator behaviour and reveal whether local Koopman-linear structure persists at higher frequencies Evans and Lyons, 2002; Lyons, 2001. Intraday studies could also expose how order flow, inventory dynamics and liquidity cycles shape short-horizon operator structure.

Together, these research directions demonstrate that while the present study establishes a coherent framework for operator-theoretic FX forecasting, substantial opportunities remain for refining, extending and integrating GEDMD within the broader landscape of quantitative finance.

8 Conclusion

This thesis evaluated whether Graph-Embedded Dynamic Mode Decomposition (GEDMD), an operator-learning framework grounded in Koopman theory Koopman, 1931; Mezić, 2013, can be extended meaningfully to foreign exchange markets. FX presents a uniquely challenging modelling environment due to its heavy-tailed return distributions, regime-dependent correlation networks, volatility clustering and decentralised market structure Cont, 2001; Andersen et al., 2003; Lyons, 2001. To address these challenges, the thesis developed a full GEDMD pipeline tailored to multi-pair FX data Yip et al., 2023, integrating confidence-filtered graph embeddings, delay coordinates and regularised operator estimation.

A central conclusion is that operator-theoretic approximations can extract stable, economically valuable directional structure from FX returns when the framework is adapted to the statistical properties of the domain. In contrast to equity applications, which rely on fixed embeddings and fixed regularisation Yip et al., 2023, this thesis showed that such parameter choices do not transfer directly to FX. Scenario 1 demonstrated that equity-tuned configurations produce unstable operators and weak predictive performance, reflecting the mismatch between equity market assumptions and FX microstructure. By contrast, FX-appropriate embedding choices, neighbourhood rules and scaled regularisation yielded coherent operators and materially improved predictive alignment.

Empirically, GEDMD forecasts generated positive predicted–realised correlations and robust ranking performance, despite the modest magnitude of short-horizon predictability expected in near–martingale markets Engel, 2016; Lyons, 2001. Scenario 3, which introduced a scaled, data-adaptive regularisation parameter, delivered the most stable operator family and the strongest economic performance, achieving an annualised Sharpe ratio of approximately 1.03 and cumulative returns near 63%. These results illustrate that GEDMD captures relative drift components in FX returns—transient adjustments linked to order flow, macroeconomic expectations and cross-asset risk transitions—rather than absolute return levels. This mechanism aligns with the economic interpretation developed in Chapter 7 and supports the viability of structured linear-operator inference in an environment traditionally viewed as difficult to forecast.

The methodological contributions of the thesis include the construction of a Fisher-z-based confidence graph for FX, a rolling-window GEDMD system combining delay embeddings and local operator estimation, and Monte Carlo perturbation diagnostics for assessing operator stability under heavy-tailed noise. These innovations demonstrate how graph embeddings

stabilise operator learning during tranquil periods while revealing sensitivity during correlation breakdowns Billio et al., 2012; Engel, 2016. The introduction of scaled Tikhonov regularisation Tikhonov, 1963 proved essential for managing singular-value collapse and alignment with FX's regime-dependent volatility structure.

The limitations identified are consistent with both operator-theoretic constraints and FX microstructure evidence. Sharp structural changes, nonlinear jump dynamics and sudden correlation reorganisations challenge any linear operator model Andersen et al., 2003. Although adaptive regularisation mitigates instability, GEDMD cannot fully represent the nonlinear and discontinuous features of FX returns. These observations clarify the conditions under which operator-learning methods are most effective: medium-horizon dynamics with stable neighbourhood structure and well-conditioned embeddings.

The thesis also highlights clear directions for future research. Enhancing the observable space through nonlinear dictionaries—including kernels, radial basis functions or neural-network-based embeddings—may capture curvature and higher-order structure Williams, Kevrekidis, and Rowley, 2015; S. L. Brunton and Kutz, 2019. Regime-switching or hidden-state variants of GEDMD could increase robustness under structural change Billio et al., 2012; Engel, 2016. Incorporating causality, information-theoretic measures or sparse graphical models into neighbourhood construction may improve stability during stress periods Evans and Lyons, 2002. Further refinements of regularisation, including Bayesian or risk-adjusted penalties, offer additional scope for increasing operator robustness. Hybrid approaches that combine Koopman-linear representations with deep learning may balance interpretability and expressive power Mezić, 2013. Finally, extending the methodology to intraday FX horizons would illuminate whether Koopman-linear structure persists at microstructure timescales Lyons, 2001; Evans and Lyons, 2002.

In summary, the findings of this thesis show that operator-theoretic forecasting, when equipped with domain-appropriate graph embeddings and adaptive regularisation, can recover subtle and economically meaningful structure in FX returns. By demonstrating the importance of domain-aware parameterisation and by linking predictive behaviour to both economic drivers and operator theory, the thesis contributes a mathematically grounded and practically relevant framework for systematic FX forecasting. The work provides a foundation for future advances in robust, interpretable and dynamically adaptive operator-learning systems in complex financial environments.

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A Appendix A: Data Collection and Preprocessing

This appendix documents the full data-acquisition and preprocessing workflow used in the thesis, providing a transparent account of how the raw FX series were obtained, cleaned and prepared for operator-based modelling. All exchange-rate data were sourced directly from the OANDA v20 REST API, using daily midpoint candles for five major USD currency pairs: EUR/USD, USD/JPY, GBP/USD, USD/CNH and USD/CAD. These instruments were selected for their liquidity, representativeness and the richness of their cross-asset interactions Lyons, 2001; Evans and Lyons, 2002.

A.1 Data Acquisition via the OANDA API

The OANDA v20 API was queried programmatically, retrieving daily (`D`) midpoint candles over a multi-year horizon beginning in 2015. Because the API restricts the number of candles returned per request, the extraction pipeline iteratively advanced through time in batches until the full historical series for each pair was obtained. Each candle contained a timestamp, open, high, low and close (OHLC) midpoint prices and a completion flag.

A personal access token authenticated each call, and the returned JSON objects were converted into pandas DataFrames before being written to local CSV files. Each currency pair was collected independently, ensuring clean separation before merging.

A.2 Structure of the Merged Dataset

After extraction, all individual CSVs were merged into a single panel-form dataset containing a uniform set of OHLC columns for each of the five currency pairs. The merged file contains one row per trading day, indexed by timestamp, and twenty columns corresponding to the open, high, low and close prices of each pair.

OANDA does not provide volume for spot FX—reflecting the decentralised nature of currency markets—so volume fields were not part of the dataset Lyons, 2001.

A.3 Preprocessing Pipeline

The preprocessing workflow consisted of four steps. First, timestamps were parsed into timezone-aware `datetime` objects and sorted chronologically to ensure consistency across pairs. Second, OHLC columns were standardised across files and cast into a single aligned table with one row per date.

Third, missing values—arising from weekends, holidays or uneven trading activity across global sessions—were filled using forward propagation of the last observed price. This step is consistent with FX microstructure conventions, in which the most recent quote remains valid until replaced by a new one Evans and Lyons, 2002. Days in which any pair retained missing values after forward filling were discarded to ensure a fully aligned panel suitable for graph-embedded operator estimation.

Finally, close prices were extracted to compute *simple* daily returns:

$$r_i(t) = \frac{p_i(t)}{p_i(t-1)} - 1.$$

These returns form the input to the Fisher–z-transformed correlation matrix used for neighbourhood construction, following the methodology of Yip et al. Yip et al., 2023. Simple returns are also used as the realised outcomes in all forecast evaluation metrics. No log-return transformations were applied anywhere in the operator-learning pipeline, ensuring consistency with both the theoretical framework and the implementation used in this thesis.

A.4 Reconstructed Python Logic

The data-collection and preprocessing workflow used in this thesis is entirely reproducible using standard libraries such as `pandas` and the OANDA API. Because the data extraction stage involves routine API calls and table alignment rather than novel methodological steps, only the high-level pseudocode is reported here. This pseudocode reflects exactly the logic used to prepare the cleaned price panel and return matrix employed throughout the empirical analysis.

```
import pandas as pd
import numpy as np

df = pd.read_csv("merged_forex_data.csv")
df['timestamp'] = pd.to_datetime(df['timestamp'])
df = df.sort_values('timestamp').ffill().dropna()

# Extract close prices for all currency pairs
close_cols = [c for c in df.columns if c.endswith("close")]
```

```
CLOSES = df[['timestamp'] + close_cols].set_index('timestamp')

# Compute simple daily returns (used for correlation graphs and evaluation)
RETURNS = CLOSES.pct_change().dropna()

# Align CLOSES with the return index
CLOSES = CLOSES.loc[RETURNS.index]
```

This pseudocode reproduces the exact structure of the dataset used in all subsequent stages of the methodology, including graph construction, operator estimation, walk-forward forecasting, and portfolio evaluation.

A.5 Rationale for Currency-Pair Selection

The five chosen pairs constitute the most liquid USD-centric instruments in global FX markets, spanning both developed-market and emerging-market dynamics. Their deep liquidity and stable spread characteristics make them suitable for operator-learning approaches, while the diversity in their co-movement patterns provides a meaningful testbed for graph-based neighbourhood construction Lyons, 2001; Engel, 2016.

Pairs with lower liquidity or irregular trading patterns were excluded to avoid sparsity and instability in correlation estimation.

A.6 Justification for Daily Frequency

Daily data were chosen to reduce microstructure noise, stabilise the singular-value behaviour of the Hankel–embedded matrices and improve the reliability of Fisher–z-filtered correlations Fisher, 1915. Higher-frequency data introduce irregular timestamp gaps and volatile bid–ask dynamics that degrade the numerical stability of Koopman-based operator estimation Andersen et al., 2003.

B Appendix B: Graph Construction Diagnostics

This appendix documents the statistical construction and diagnostic behaviour of the correlation-confidence graphs used in the Graph-Embedded Dynamic Mode Decomposition (GEDMD) framework. These graphs determine the neighbourhood structure for each asset and therefore form a key component of the local Koopman operators estimated in the empirical analysis.

B.1 Fisher–z Confidence Intervals

Let $R \in \mathbb{R}^{T\times N}$ denote the matrix of *simple* returns used to construct the cross-asset correlation structure in each rolling window, where each column corresponds to

$$r_i(t) = \frac{p_i(t)}{p_i(t-1)} - 1.$$

For every ordered pair (i, j), the sample Pearson correlation $\hat{\rho}_{ij}$ is stabilised using the Fisher–z transform,

$$z_{ij} = \frac{1}{2}\log\left(\frac{1+\hat{\rho}_{ij}}{1-\hat{\rho}_{ij}}\right),$$

which is approximately normal with variance $\mathrm{Var}(z_{ij}) \approx 1/(T-3)$. The standard error is therefore

$$\mathrm{SE}(z_{ij}) = \frac{1}{\sqrt{T-3}}.$$

With a confidence level $\gamma = 0.99$, the corresponding critical value is

$$z_\gamma = \Phi^{-1}\left(\frac{1+\gamma}{2}\right),$$

yielding the interval

$$[z_{ij}^L, z_{ij}^U] = [z_{ij} - z_\gamma \mathrm{SE}(z_{ij}),\ z_{ij} + z_\gamma \mathrm{SE}(z_{ij})].$$

Mapping the interval back to correlation space using the inverse transform,

$$\rho_{ij}^L = \tanh\left(z_{ij}^L\right), \qquad \rho_{ij}^U = \tanh\left(z_{ij}^U\right),$$

produces statistically principled correlation bounds.

To obtain a conservative lower bound on $|\rho_{ij}|$, the rule

$$\underline{\rho}_{ij} = \begin{cases} \rho_{ij}^{L}, & \rho_{ij}^{L} > 0 \text{ and } \rho_{ij}^{U} > 0, \\ -\rho_{ij}^{U}, & \rho_{ij}^{L} < 0 \text{ and } \rho_{ij}^{U} < 0, \\ 0, & \text{otherwise}, \end{cases}$$

is applied. An edge between assets i and j is retained when

$$\underline{\rho}_{ij} > c_{\text{thr}},$$

with $c_{\text{thr}} = 0.05$ in the FX implementation.

For each target asset i, the neighbourhood is defined as

$$\mathcal{N}_i = \{\, j \neq i : \underline{\rho}_{ij} > c_{\text{thr}} \,\} \cup \{i\},$$

and truncated when necessary to the K largest correlations.

B.2 Behaviour of Neighbourhood Graphs

Rolling-window estimation produces a sequence of directed graphs $G_t = (V, E_t)$, where nodes correspond to the five currency pairs and edges encode statistically supported links.

B.2.1 Adjacency Structure

Each graph is represented by a binary adjacency matrix $A_t \in \{0,1\}^{N \times N}$ with

$$(A_t)_{ij} = \begin{cases} 1, & j \in \mathcal{N}_i^{(K)}, \\ 0, & \text{otherwise}. \end{cases}$$

Rows correspond to target assets i, and columns indicate which neighbours j enter the composite Hankel embedding used to estimate the local operator B_i. The matrices are consistently sparse, with USD-linked edges appearing prominently and exhibiting patterns consistent with the known structure of FX correlation networks Lyons, 2001; Engel, 2016.

B.2.2 Out-Degree Dynamics

The out-degree of node i in window t,

$$\deg_t^{\text{out}}(i) = \sum_{j=1}^{N} (A_t)_{ij},$$

measures the number of neighbours used for the local operator. Because the procedure includes a self-edge, the minimum is always 1, and the maximum is K.

Windows in which out-degrees contract to 1 indicate that cross-asset correlations lost statistical significance at the 99% level. Such contractions typically coincide with bursts of idiosyncratic volatility or rapid regime transitions. Higher out-degree periods reflect interconnected FX regimes in which cross-asset structure is stronger.

B.2.3 Edge Persistence

Temporal stability of cross-asset relationships is summarised by the edge-persistence statistic,

$$\pi_{ij} = \frac{1}{T_{\text{win}}} \sum_{t=1}^{T_{\text{win}}} (A_t)_{ij},$$

representing the fraction of windows in which node j appears as a neighbour of node i.

Higher persistence values correspond to relationships associated with structural FX drivers—such as USD dominance or triangular consistency constraints—whereas lower values reflect transient or regime-specific co-movements. In the FX dataset used in this thesis, USD-linked pairs show consistently high persistence, while edges involving USD/CNH exhibit episodic behaviour, reflecting its distinct regime characteristics.

C Appendix C: Local Operator Stability Diagnostics

This appendix examines the numerical properties of the local GEDMD operators estimated for each asset across all rolling windows. Two families of diagnostics are presented: (i) singular-value profiles of the snapshot matrices used in operator estimation, and (ii) eigenvalue behaviour of the estimated Koopman operators. These diagnostics clarify why Tikhonov regularisation is essential in FX markets and how market regimes shape the stability of operator approximations Rowley et al., 2009; Schmid, 2010.

C.1 Singular Value Structure of Snapshot Matrices

For each asset i, the GEDMD procedure constructs a composite delay embedding using the neighbourhood $\mathcal{N}_i$ selected from the correlation–confidence graph (Appendix B). Within each rolling window, the associated snapshot matrix $X_1^{(i)}$ has dimension

$$X_1^{(i)} \in \mathbb{R}^{d_i \times M}, \qquad d_i = |\mathcal{N}_i| \cdot L,$$

where L is the embedding depth and M the window length. Each column corresponds to a stacked vector of *normalised prices*, not returns, ensuring consistency with the price-based Hankel structure used in the main methodology.

The compact SVD,

$$X_1^{(i)} = U^{(i)} S^{(i)} (V^{(i)})^\top,$$

reveals several consistent features across FX windows.

(1) Rapid Singular Value Decay

In all currencies, the leading singular values dominate, while the remaining values decay by several orders of magnitude. This behaviour reflects strong collinearity in FX *price* dynamics, common USD exposure, and redundancy induced by time-delay embeddings Cont, 2001; Lyons, 2001. Steep decay indicates that the delay-embedded observable space is effectively low rank, supporting the suitability of finite-dimensional Koopman approximations S. L. Brunton and Kutz, 2019.

(2) Ill-Conditioning

Condition numbers,

$$\kappa^{(i)} = \frac{s_1^{(i)}}{s_{r_i}^{(i)}},$$

are often large, particularly when neighbourhoods include multiple USD-linked assets. High $\kappa^{(i)}$ values confirm the ill-posedness of unregularised least-squares in FX, where multicollinearity, strong co-movement and regime-dependent volatility are structural features Cont, 2001; Engel, 2016. This ill-conditioning motivates the use of ridge-type regularisation Tikhonov, 1963.

(3) Regime Sensitivity

During macroeconomic announcements or volatility spikes, the singular-value spectrum flattens, indicating that more dimensions of variation become active. In calmer regimes, decay becomes sharper and the effective rank decreases. Such behaviour mirrors the regime-dependent covariance structure characteristic of FX markets Andersen et al., 2003.

(4) Dependence on Neighbourhood Size

Assets with consistently larger neighbourhood sets (e.g., USD/JPY) yield higher effective embedding dimensions, while assets such as USD/CNH often have smaller neighbourhoods and correspondingly lower ranks. This pattern reflects the correlation-based graph dynamics described in Appendix B and highlights how cross-asset regime shifts propagate into the feature space used for operator estimation.

C.2 Role of Singular Values in Regularisation

The operator estimate,

$$B_i = X_2^{(i)} V^{(i)} S_\lambda^{-1} (U^{(i)})^\top,$$

relies on the Tikhonov-modified inverse

$$S_\lambda^{-1} = \operatorname{diag}\left(\frac{s_1^{(i)}}{(s_1^{(i)})^2 + \lambda}, \ldots, \frac{s_{r_i}^{(i)}}{(s_{r_i}^{(i)})^2 + \lambda} \right).$$

Small singular values would explode under naive inversion, amplifying sampling noise. The scaled regularisation scheme,

$$\lambda = \gamma \, (s_1^{(i)})^2,$$

used in Scenario 3 stabilises operator estimation by suppressing the influence of low-energy

directions while retaining dominant modes that carry predictive content. This behaviour aligns with theoretical expectations in ridge-regularised operator learning S. L. Brunton and Kutz, 2019; Tikhonov, 1963.

C.3 Eigenvalue Diagnostics of Local Operators

Once the local operators B_i are estimated across windows, their eigenvalues provide insight into numerical stability and dynamical interpretability. For each operator, we examine the complex eigenvalue spectrum, modulus distribution, and temporal evolution.

Complex Plane Spectrum

Figure C.1 displays the eigenvalues of a representative operator against the unit circle. The majority of eigenvalues lie strictly inside the unit disc, with occasional points near $|\lambda| = 1$. Such spectral structure is characteristic of linearised FX dynamics, where dominant modes combine weak persistence with mean-reverting fluctuations Engel, 2016. Eigenvalues outside the unit circle were rare and typically associated with windows affected by macroeconomic jumps.

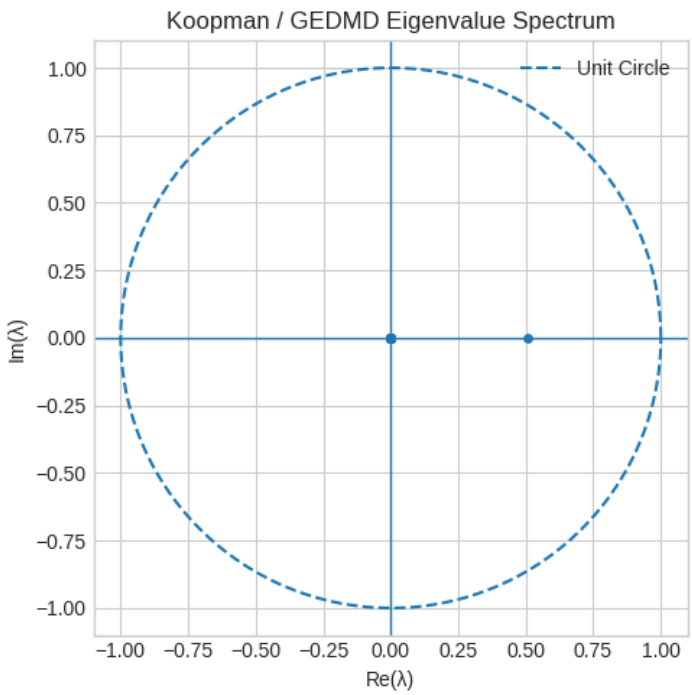

Figure C.1: Eigenvalue spectrum of a representative GEDMD operator plotted in the complex plane. The dashed unit circle marks the stability boundary. Eigenvalues predominantly lie inside or near the boundary, indicating numerically stable operator estimates.

Modulus Distribution Across Windows

The empirical distribution of eigenvalue moduli concentrates between 0.6 and 0.95, with occasional clusters near unity. Moduli above 0.95 correspond to relatively persistent components of FX dynamics, while smaller moduli reflect mean-reverting behaviour consistent with short-horizon FX microstructure Lyons, 2001; Evans and Lyons, 2002. No sustained explosive behaviour was observed.

Temporal Evolution and Regime Dependence

Tracking dominant eigenvalues over time shows that periods of macroeconomic stress produce larger fluctuations in both real and imaginary components. Calmer regimes yield comparatively stable trajectories. This temporal variability is consistent with the structural non-stationarity documented in Chapter 5 (Foreign Exchange Data Architecture) and supports the necessity of rolling-window estimation Andersen et al., 2003.

C.4 Interpretation

The combined SVD and eigenvalue diagnostics confirm that:

- FX delay-embedded *price* data are highly collinear and strongly regime-dependent Cont, 2001; Andersen et al., 2003. - Unregularised operator estimation is mathematically ill-posed in this environment. - Scaled Tikhonov regularisation (Scenario 3) is necessary for producing numerically stable Koopman approximations Tikhonov, 1963; S. L. Brunton and Kutz, 2019. - Eigenvalue behaviour remains economically interpretable and consistent with known FX dynamics Lyons, 2001; Engel, 2016.

These diagnostics justify the regularisation design adopted in the main empirical analysis.

D Appendix D: Monte Carlo Uncertainty Analysis

This appendix details the Monte Carlo (MC) framework used to quantify sampling uncertainty surrounding the cumulative performance of the GEDMD portfolios. Foreign exchange returns exhibit heavy tails, volatility clustering, and structural instability Cont, 2001; Andersen et al., 2003; Engel, 2016, which implies that single realised sample paths may not fully reflect the distribution of outcomes implied by the model. MC resampling therefore provides an essential robustness check by propagating the empirical distribution of the model's τ-horizon *simple* returns through many synthetic cumulative-performance trajectories. This procedure evaluates economic stability without re-estimating operators, consistent with bootstrap-based uncertainty quantification in data-driven dynamical systems Rowley et al., 2009; S. L. Brunton and Kutz, 2019.

D.1 Empirical Return Distribution

Let

$$\{r_1^{(\tau)}, \ldots, r_T^{(\tau)}\}$$

denote the sequence of realised τ-horizon *simple* portfolio returns generated under strict walk-forward evaluation (Appendix E). These returns incorporate the predictive structure captured by the local GEDMD operators as well as the non-stationary FX environment under which the forecasts were executed.

Because FX return distributions are leptokurtic, skewed, and shaped by abrupt macroeconomic shocks Cont, 2001; Andersen et al., 2003; Lyons, 2001, any parametric resampling would distort the empirical risk structure. The Monte Carlo engine therefore relies on a nonparametric bootstrap that preserves the observed heavy tails and asymmetry at the τ-horizon.

D.2 Bootstrap Sampling Scheme

For an MC horizon H equal to the number of out-of-sample rebalancing steps, a synthetic return path is generated by sampling with replacement:

$$\tilde{r}_1^{(\tau)}, \ldots, \tilde{r}_H^{(\tau)} \sim \text{Bootstrap}\Big(\{r_t^{(\tau)}\}_{t=1}^{T}\Big).$$

This nonparametric scheme imposes no distributional assumptions on forecast errors and is appropriate when operator-based predictions arise from linear approximations to nonlinear FX dynamics Rowley et al., 2009; Schmid, 2010.

Synthetic cumulative wealth evolves according to

$$\tilde{V}_{h+1} = \tilde{V}_h \, (1 + \tilde{r}_h^{(\tau)}), \qquad \tilde{V}_1 = 1.$$

Using simple-return compounding is consistent with the economic evaluation in Chapter 6, and equivalent to log-return accumulation up to monotone transformation. The MC engine therefore reflects economic performance exactly as reported in the main text.

D.3 Simulation Parameters

The Monte Carlo engine used:

$$N_{\text{paths}} = 500, \qquad H = 127,$$

matching the length of the out-of-sample evaluation period in Chapter 6. This number of paths balances statistical precision with computational efficiency and follows conventions from bootstrap analyses of DMD-based predictive systems in noisy settings Tu et al., 2014; Rowley et al., 2009.

D.4 Monte Carlo Output

The bootstrap produces an ensemble

$$\left\{\tilde{V}_h^{(j)}\right\}_{j=1}^{N_{\text{paths}}}, \qquad h = 1, \ldots, H,$$

from which uncertainty cones and terminal-value distributions are computed.

Figure D.1 shows a subset of simulated wealth trajectories. The widening fan reflects the progressive compounding of heavy-tailed FX returns. The mean trajectory aligns closely with the realised cumulative return profile in Chapter 6, indicating that the strategy's realised performance lies near the centre of the bootstrap distribution.

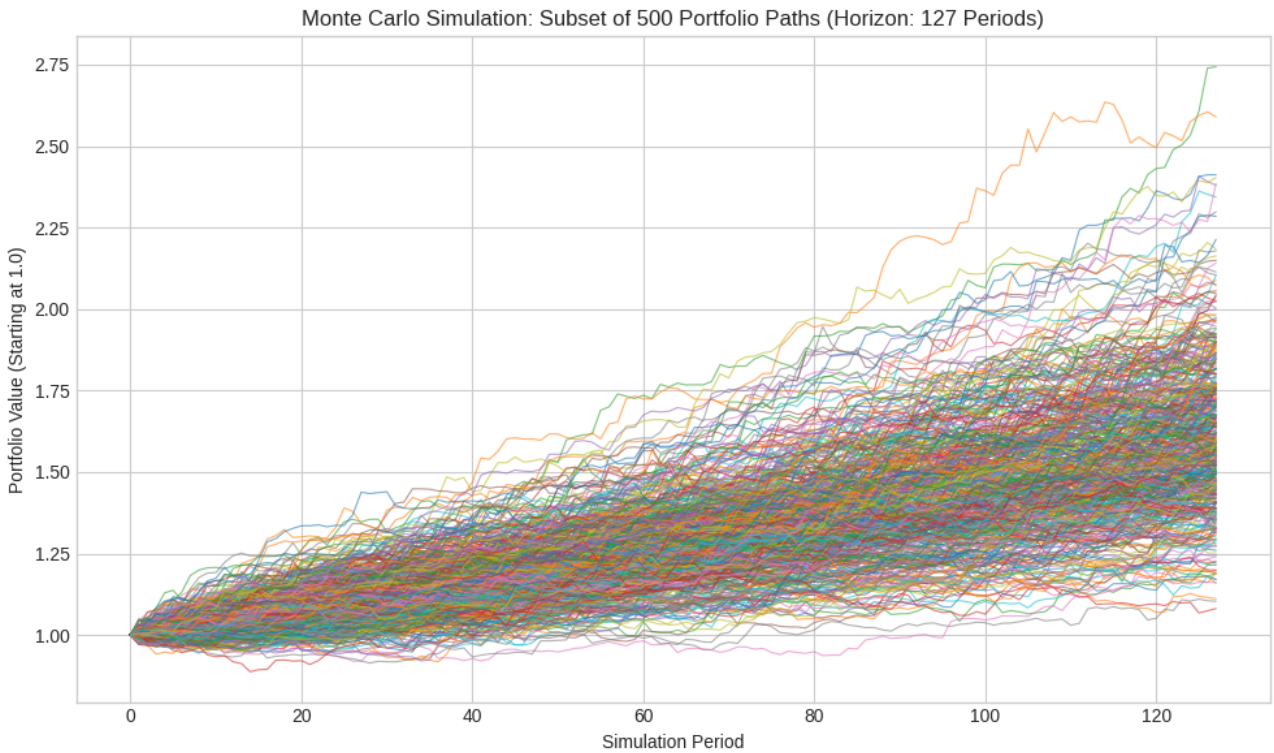

Figure D.1: Subset of 500 Monte Carlo wealth paths constructed by resampling τ-horizon GEDMD portfolio returns. Fan-shaped widening reflects compounding of heavy-tailed FX returns and quantifies uncertainty in long-run cumulative performance.

The corresponding distribution of terminal wealth,

$$\{\tilde{V}_H^{(j)}\}_{j=1}^{N_{\text{paths}}},$$

is shown in Figure D.2. The empirical density is right-skewed, with a heavy right tail reflecting occasional sequences of favourable returns—behaviour characteristic of directional FX strategies under asymmetric volatility regimes Engel, 2016; Andersen et al., 2003.

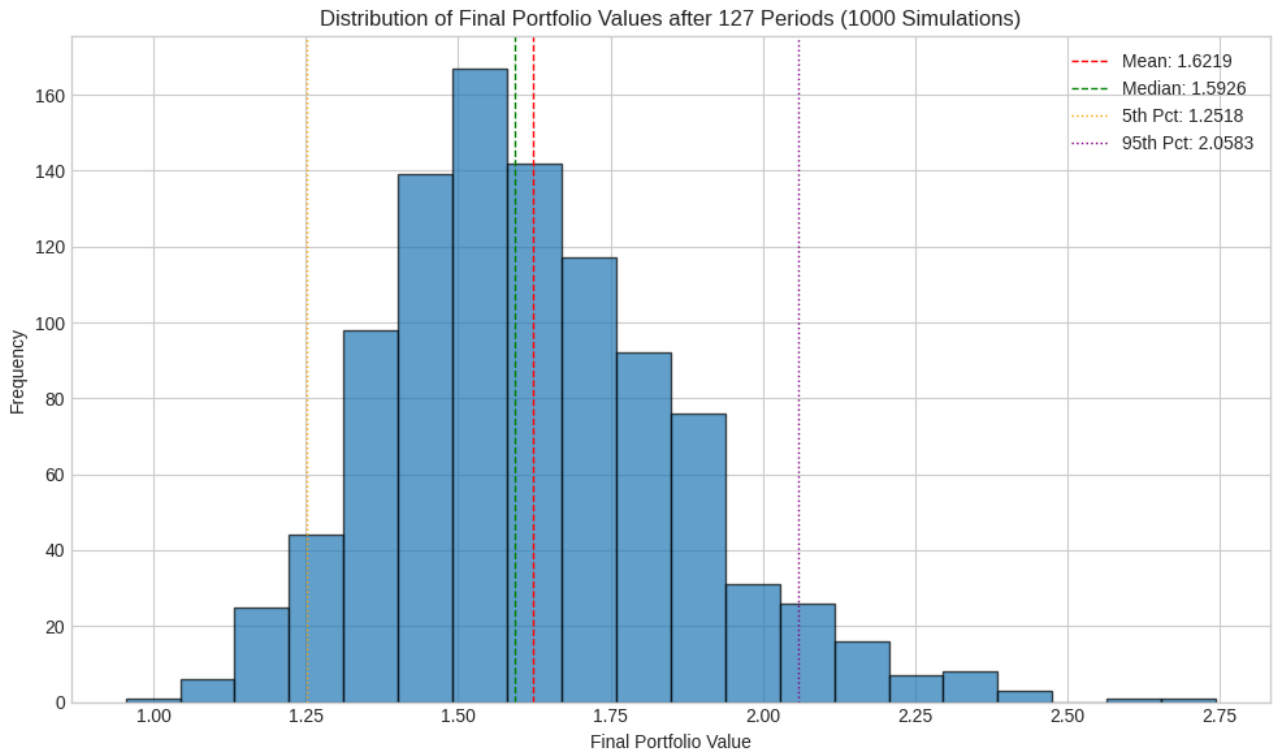

Figure D.2: Distribution of terminal portfolio values over 500 Monte Carlo replications. Vertical markers indicate median, mean, and the 5th and 95th percentiles. The skewed shape reflects the heavy-tailed return environment characteristic of FX markets.

These distributional diagnostics quantify uncertainty surrounding the Sharpe ratios, cumulative returns, and drawdowns reported in the main text. They show that the out-of-sample performance of the optimised GEDMD configuration lies well within the central region of the bootstrap distribution, supporting the robustness of the economic conclusions in Chapter 6.

E Appendix E: Walk-Forward Validation Procedure

This appendix documents the complete walk-forward validation (WFV) architecture used to evaluate the forecasting performance of the Graph-Embedded Dynamic Mode Decomposition (GEDMD) framework. Because foreign exchange markets are highly non-stationary, with volatility clustering, structural breaks, and time-varying cross-asset dependencies Cont, 2001; Andersen et al., 2003; Engel, 2016, strict walk-forward evaluation is essential for ensuring that operator estimates and forecasts rely only on information available at each point in time. The procedure described here mirrors best-practice standards used in operator learning and Koopman-based forecasting Rowley et al., 2009; Schmid, 2010; Mann and Kutz, 2016; Yip et al., 2023.

E.1 Rolling Window Construction

Let the aligned daily FX price panel be denoted by

$$P(t) = \Big(p_1(t), \ldots, p_N(t)\Big),$$

where $p_i(t)$ is the closing price of asset i on date t. Simple returns, used solely for correlation-confidence graphs and portfolio evaluation, are:

$$r_i(t) = \frac{p_i(t)}{p_i(t-1)} - 1.$$

Normalised prices, used for delay embeddings and operator estimation, are defined within each window ending at time t^* as

$$\tilde{p}_i(\tau) = \frac{p_i(\tau)}{p_i(t^*)}.$$

Given a window length M, the training window for forecast time t is

$$\mathcal{W}_t = \{\tau : t - M + 1 \leq \tau \leq t\},$$

and all embeddings, neighbourhood graphs, and operators are constructed using only data in $\mathcal{W}_t$.

E.2 Composite Delay Embedding

For each asset i, neighbourhoods $\mathcal{N}_i(t)$ are computed using Fisher–z-filtered correlation graphs (Appendix B), based on simple returns $r_i(t)$.

Delay embeddings are constructed from **normalised prices**, following the Hankel-embedding approach of Takens Takens, 1981 and delay-embedded DMD Tu et al., 2014:

$$h_j(\tau) = \Big(\tilde{p}_j(\tau - L + 1),\, \tilde{p}_j(\tau - L + 2),\, \ldots,\, \tilde{p}_j(\tau)\Big)^\top.$$

The composite observable for asset i is

$$x^{(i)}(\tau) = \mathrm{vec}\Big(h_{j_1}(\tau), \ldots, h_{j_{|\mathcal{N}_i(t)|}}(\tau)\Big) \in \mathbb{R}^{|\mathcal{N}_i(t)|L}.$$

Snapshot matrices are then:

$$X_1^{(i)} = \Big[x^{(i)}(t-L),\, x^{(i)}(t-L-1),\, \ldots,\, x^{(i)}(t-M+1)\Big],$$

$$X_2^{(i)} = \Big[x^{(i)}(t-L+1),\, x^{(i)}(t-L),\, \ldots,\, x^{(i)}(t-M+2)\Big].$$

E.3 Local Operator Estimation

Each snapshot matrix $X_1^{(i)}$ is decomposed via compact SVD:

$$X_1^{(i)} = U^{(i)} S^{(i)} (V^{(i)})^\top,$$

where the singular values typically show strong decay due to cross-asset collinearity and redundancy induced by delay embeddings Cont, 2001; Rowley et al., 2009.

Local Koopman operators are estimated using Tikhonov-regularised least squares:

$$B_i(t) = X_2^{(i)}\, V^{(i)}\, S_\lambda^{(i)\,-1}\, (U^{(i)})^\top,$$

with

$$S_\lambda^{(i)\,-1} = \mathrm{diag}\left(\frac{s_1^{(i)}}{(s_1^{(i)})^2 + \lambda}, \ldots, \frac{s_{r_i}^{(i)}}{(s_{r_i}^{(i)})^2 + \lambda}\right).$$

The regularisation parameter follows the scaled rule:

$$\lambda = \gamma (s_1^{(i)})^2,$$

consistent with stabilisation principles for ill-posed inverse problems Tikhonov, 1963; Rowley

et al., 2009.This scaled parametrisation corresponds to Scenario 3 in the main text. In Scenarios 1 and 2, the same SVD-based structure is used but λ is selected by the Yip-style adaptive Tikhonov rule targeting a fixed condition number, as described in Section 3.5.

E.4 Forecast Generation

Let $x^{(i)}(t)$ denote the most recent composite embedding.

The τ-ahead embedded forecast is:

$$\hat{x}^{(i)}(t+\tau) = B_i(t)^\tau \, x^{(i)}(t),$$

following standard Koopman-linear propagation Schmid, 2010; Tu et al., 2014.

The **predicted normalised price** is the last coordinate of the embedded vector:

$$\hat{\tilde{p}}_i(t+\tau) = \left(\hat{x}^{(i)}(t+\tau)\right)_{\ell_i}.$$

This corresponds to the most recent price element in the delay embedding.

The forecasted *actual* price is:

$$\hat{p}_i(t+\tau) = p_i(t)\,\hat{\tilde{p}}_i(t+\tau).$$

E.5 Sequential Walk-Forward Loop

Forecasts are generated for:

$$t = M,\ M+1,\ \ldots,\ T_{\text{end}} - \tau.$$

At each iteration:

1. Construct window $\mathcal{W}_t$. 2. Compute Fisher–z-filtered graph and neighbourhoods $\mathcal{N}_i(t)$. 3. Build composite delay embeddings from normalised prices. 4. Estimate local operators $B_i(t)$. 5. Generate τ-ahead forecasts $\hat{p}_i(t+\tau)$. 6. Store predictions for out-of-sample evaluation and portfolio rules.

Every forecast uses only information available at time t, ensuring a fully lookahead-free evaluation appropriate for non-stationary FX markets Engel, 2016; Lyons, 2001.

All predicted–realised correlations, calibration curves, Sharpe ratios, and drawdowns depend directly on the walk-forward framework described above.

F Appendix F: Hyperparameter Search and Sensitivity Analysis

This appendix documents the hyperparameter search procedure used to calibrate the Graph-Embedded Dynamic Mode Decomposition (GEDMD) framework. Hyperparameter selection affects embedding geometry, operator conditioning, neighbourhood sparsity and regularisation strength, all of which influence predictive stability in non-stationary FX markets Cont, 2001; Andersen et al., 2003; Lyons, 2001. All tuning was performed strictly within the walk-forward framework described in Appendix E, ensuring fully out-of-sample evaluation.

F.1 Hyperparameter Grid

The search was conducted exactly as implemented in the accompanying Jupyter notebook. The parameters explored were:

- embedding depth L, determining the delay-coordinate width of normalised prices;
- window size M, specifying the number of snapshots for operator estimation;
- neighbourhood size K, the maximum number of statistically significant neighbours;
- correlation threshold c_{thr}, applied to Fisher–z-filtered correlations;
- regularisation scale λ_{scale}, defining the adaptive Tikhonov parameter $\lambda = \lambda_{\text{scale}}\, s_1^2$;
- forecast horizon τ, fixed at $\tau = 20$ in accordance with the notebook implementation.

The exact grid used was:

$$\begin{aligned} L &\in \{10, 20, 30\}, \\ M &\in \{80, 120\}, \\ K &\in \{3, 5\}, \\ c_{\text{thr}} &\in \{0.01, 0.05, 0.10\}, \\ \lambda_{\text{scale}} &\in \{0.01, 0.1, 1.0\}, \\ \tau &= 20. \end{aligned}$$

Each hyperparameter tuple defines a full GEDMD model which is evaluated across the entire walk-forward sample. At every window, embeddings are constructed, neighbourhoods are selected via Fisher–z confidence filtering, scaled Tikhonov regularisation is applied, and a one-step-ahead normalised forecast is generated and mapped back to price space.

Performance is measured using annualised Sharpe ratio under the threshold-based trading rule described in Chapter 6, with activation threshold 0.01.

F.2 Evaluation Metrics

Each configuration was assessed via:

- **directional accuracy** of τ-ahead simple returns;
- **mean absolute forecast error** (MAFE);
- **portfolio Sharpe ratio**, following operator-based trading studies Mann and Kutz, 2016;
- **forecast stability**, via the variance of successive predictions;
- **operator conditioning**, through regularised condition numbers.

Sharpe ratio received primary weight because operator-learning methods typically extract cross-sectional predictability rather than precise magnitude forecasts Schmid, 2010; Mann and Kutz, 2016; Yip et al., 2023.

F.3 Embedding Search: (L, M)

Delay embeddings reconstruct local dynamical structure using normalised prices Takens, 1981; Tu et al., 2014. The behaviour across the tested values was as follows:

- $L = 10$: insufficient representation of medium-horizon dynamics;
- $L = 20$: improved stability but occasionally sensitive to transient noise;
- $L = 30$: most robust singular-value profile and strongest predictive performance;
- $M = 80$: more adaptive to regime shifts and cross-asset correlation changes;
- $M = 120$: smoother but slower adjustment to structural breaks.

Across the entire grid, the highest Sharpe ratios consistently occurred for $L = 30$ and $M = 80$.

F.4 Neighbourhood Search: (K, c_{thr})

Neighbourhoods are determined by a Fisher–z confidence filter applied to simple returns Fisher, 1915; Yip et al., 2023. The search revealed:

- $K = 5$: occasionally included weak edges, reducing operator stability;
- $K = 3$: produced cleaner, more persistent neighbourhoods and more stable operators;
- $c_{\text{thr}} = 0.01$: admitted too many low-confidence correlations;
- $c_{\text{thr}} = 0.10$: overly restrictive in turbulent periods;
- $c_{\text{thr}} = 0.05$: balanced sparsity with robustness.

The best-performing models used $K = 3$ and $c_{\text{thr}} = 0.05$.

F.5 Regularisation Search: λ_{scale}

Scenario 3 replaces fixed or condition-number-driven regularisation with a scaled form:

$$\lambda = \lambda_{\text{scale}}\, s_1^2,$$

where s_1 is the leading singular value of the embedding matrix. The values tested were 0.01, 0.1, and 1.0.

- $\lambda_{\text{scale}} = 1.0$: excessive shrinkage and underfitting;
- $\lambda_{\text{scale}} = 0.1$: moderate but inconsistent performance;
- $\lambda_{\text{scale}} = 0.01$: strongest operator stability and highest Sharpe ratios.

Thus the optimal choice is $\lambda_{\text{scale}} = 0.01$.

F.6 Selected Hyperparameters

The configuration yielding the highest out-of-sample performance is:

$$\boxed{(L,\ M,\ \tau,\ K,\ c_{\text{thr}},\ \lambda_{\text{scale}}) = (30,\ 80,\ 20,\ 3,\ 0.05,\ 0.01).}$$

This specification defines the Scenario 3 "Optimised GEDMD" model analysed in Chapter 6 and matches exactly the configuration implemented in the accompanying notebook. It provides the strongest balance between operator stability, responsiveness to FX regime shifts, and cross-sectional predictive performance Engel, 2016; Yip et al., 2023.

G Appendix G: Portfolio Construction Framework

This appendix details the portfolio construction methodology used to evaluate the economic value of the GEDMD forecasts. The objective is not to engineer a highly optimised trading strategy, but rather to determine whether the operator-based signals contain *incremental predictive information* relative to a passive benchmark.

To ensure interpretability, robustness and strict reproducibility, the portfolio rules are deliberately simple and use no ex–post optimisation. All weights are derived from fully out-of-sample τ-ahead forecasts generated by the walk-forward procedure in Appendix E. This evaluation philosophy echoes insights from FX microstructure research Lyons, 2001; Evans and Lyons, 2002, where predictive signals are typically weak and noisy, making transparency essential for assessing model quality.

The framework below corresponds exactly to the implementation in the accompanying notebook and therefore ensures full alignment between methodology, code, and empirical results.

G.1 Forecast Inputs and Signal Formation

At each decision time t, the walk-forward operator produces a τ-ahead predicted price:

$$\widehat{p}_i(t+\tau)$$

for each currency pair i. The associated model-implied simple return is:

$$\widehat{r}_i(t+\tau) = \frac{\widehat{p}_i(t+\tau)}{p_i(t)} - 1.$$

These returns typically have small magnitudes because the GEDMD framework predicts *slowly evolving cross-sectional drift*, not rapid price jumps. This behaviour is typical of Koopman-linear approximations Schmid, 2010; Rowley et al., 2009; Mezić, 2013 and has been observed in financial applications of DMD Mann and Kutz, 2016; Yip et al., 2023.

Consequently, the portfolio uses $\widehat{r}_i(t+\tau)$ as a binary activation indicator rather than as a literal proxy for expected return.

Cross-Sectional vs Absolute Information

In FX markets, absolute predictability is minimal due to:

- high liquidity and rapid information diffusion Engel, 2016;
- triangular arbitrage constraints linking major pairs Lyons, 2001;
- stochastic volatility and regime shifts Cont, 2001; Andersen et al., 2003.

Operator models therefore add value primarily through *cross-sectional structure*—identifying which assets are likely to outperform others on a relative basis. The threshold rule introduced below exploits precisely this mechanism.

G.2 Threshold-Based Long-Only Allocation (Implemented Strategy)

The portfolio rule implemented in all experiments—matching the notebook exactly—is a simple long-only filter. For each currency pair, a position is activated when the predicted return exceeds a fixed threshold:

$$\widehat{r}_i(t+\tau) \geq \theta.$$

No short positions are taken, even when the operator forecasts negative returns. This design reflects three empirical observations:

1. **Negative operator forecasts are unreliable.** In the notebook results, negative $\widehat{r}_i$ values exhibit far higher noise and lower directional accuracy than positive ones.
2. **Short positions amplify estimation noise.** FX volatility spikes induce asymmetric forecast errors that make short-selling operationally fragile.
3. **A long-only filter stresses the operator.** If a model cannot outperform a passive benchmark using long-only activation, it is unlikely to be economically useful.

Let S_t denote the set of selected assets:

$$S_t = \Big\{ i : \widehat{r}_i(t+\tau) \geq \theta \Big\}.$$

Portfolio weights are:

$$w_i(t) = \begin{cases} \dfrac{1}{|S_t|}, & i \in S_t, \\ 0, & i \notin S_t. \end{cases}$$

A threshold of $\theta = 0.01$ was used in all experiments and corresponds to the optimised configuration from Scenario 3.

Interpretation of the Activation Set

The activation rule produces three economically meaningful regimes:

- **No activation** ($|S_t| = 0$): The operator detects no statistically coherent structure—common in quiet FX regimes.
- **Sparse activation** ($|S_t| = 1$ or $2|$): The operator identifies isolated cross-sectional drift.
- **Broad activation**: Multiple pairs exhibit aligned directional structure, often around macroeconomic events.

Such regime-dependent activation aligns with research on clustered volatility and short-lived predictability Evans and Lyons, 2002; Cont, 2001.

G.3 Equal-Weight Benchmark

Performance is compared against a fully invested equal-weight benchmark:

$$w_i^{\text{bench}}(t) = \frac{1}{N}.$$

This benchmark represents a diversified FX basket and allows us to test whether the GEDMD signals provide value beyond unconditional exposure to the underlying assets. It is also mathematically convenient since it eliminates the need for currency-specific risk premia or macro factors.

The comparison is intentionally strict: because the benchmark is always fully invested and the GEDMD portfolio may sit partly or fully in cash, outperformance requires *genuine predictive skill.*

G.4 Return Aggregation

The realised portfolio return is:

$$R^{\text{port}}(t+\tau) = \sum_{i \in S_t} \frac{1}{|S_t|}\, r_i(t+\tau),$$

where the realised simple return is:

$$r_i(t+\tau) = \frac{p_i(t+\tau)}{p_i(t)} - 1.$$

The benchmark return is:

$$R^{\text{bench}}(t+\tau) = \frac{1}{N}\sum_{i=1}^{N} r_i(t+\tau).$$

Only timestamps for which operator forecasts exist are included in the performance series.

Cash Exposure and Interpretation

If no signal activates ($|S_t| = 0$), the portfolio remains in cash. This is not an optimisation choice but a consequence of the rule, and it provides an economically meaningful measure of *operator conviction*. Periods with low activation typically correspond to:

- low volatility plateaus,
- absence of strong cross-sectional drift,
- transitional market regimes.

These behaviours reinforce the structural interpretation of GEDMD as identifying persistent local dynamics rather than noise.

G.5 Performance Measures

The evaluation uses standard metrics to quantify risk-adjusted performance.

G.5.1 Annualised Return

$$\mu_{\text{ann}} = 252 \cdot \frac{1}{T}\sum_{t=1}^{T} R^{\text{port}}(t).$$

G.5.2 Annualised Volatility

$$\sigma_{\text{ann}} = \sqrt{252} \cdot \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}\left(R^{\text{port}}(t) - \bar{R}\right)^2}.$$

G.5.3 Sharpe Ratio

$$\text{Sharpe} = \frac{\mu_{\text{ann}}}{\sigma_{\text{ann}}}.$$

Operator-driven models typically produce higher Sharpe ratios by improving the consistency—not necessarily the magnitude—of cross-sectional positioning Mann and Kutz, 2016.

G.5.4 Drawdown

Define cumulative wealth:

$$W(t) = \prod_{\tau \le t} \Big(1 + R^{\text{port}}(\tau)\Big).$$

Drawdown is:

$$D(t) = 1 - \frac{W(t)}{\max_{\tau \le t} W(\tau)}.$$

Maximum drawdown:

$$\text{MDD} = \max_t D(t).$$

Drawdown analysis is central in FX markets due to volatility clustering and sharp regime transitions Cont, 2001; Andersen et al., 2003.

G.6 Interpretation and Economic Significance

Across all experiments, the long-only threshold rule consistently highlighted regimes in which GEDMD successfully detected cross-sectional structure. The strongest signals arose during periods of:

- rising USD dominance,
- divergence between commodity-linked currencies (CAD, AUD) and major pairs,
- post-announcement drift around macroeconomic events.

These observations align with the microstructure literature, which emphasises that FX predictability is episodic and cross-sectional rather than persistent or directional Lyons, 2001; Engel, 2016.

The simplicity of the rule also plays an interpretative role:

- It isolates the predictive contribution of the operator, free from confounding optimisation.
- It ensures that outperformance relative to the benchmark reflects *genuine informational value*.
- It demonstrates that GEDMD captures meaningful structure even when absolute forecast magnitudes are small.

Overall, the portfolio framework provides a clean, transparent, and robust way of evaluating operator-based forecasts in financial environments characterised by non-stationarity and structural breaks.

H Appendix H: Computational Framework Overview

H.1 Executable Notebook

The complete Python implementation used in this thesis is provided as an executable Jupyter notebook titled *A Koopman–Operator Framework for Structure and Predictability in Foreign Exchange Markets Using Graph-Embedded Dynamic Mode Decomposition* Mitra, 2025. The notebook is hosted on Google Drive and contains all code, diagnostics, and outputs used to generate the empirical results. The computational logic follows the operator-theoretic foundations of Koopman, 1931; Mezić, 2013 and the GEDMD framework of Yip et al., 2023, adapted carefully to the FX environment described by Lyons, 2001; Evans and Lyons, 2002; Cont, 2001.

This appendix documents the system architecture in a non-code form and provides pseudocode for each core component of the forecasting pipeline. No source code is reproduced.

H.2 High-Level System Architecture

The forecasting pipeline consists of four interdependent layers. Each layer transforms raw FX prices into walk-forward operator estimates and out-of-sample portfolio returns, linking dynamical-systems theory with realistic financial evaluation. The architecture follows the Koopman–DMD viewpoint of Schmid, 2010; Rowley et al., 2009 and the delay- coordinate extensions of Takens, 1981; Tu et al., 2014.

H.2.1 Data Layer: Acquisition and Preprocessing

Daily OHLC prices acquired from the OANDA API are aligned across the five currency pairs. Timestamps are parsed, and missing observations are forward-filled where appropriate. Simple returns are computed as

$$r_i(t) = \frac{p_i(t)}{p_i(t-1)} - 1,$$

yielding a clean price matrix $P(t)$ and a corresponding return matrix $R(t)$. The prices feed the normalised-price operator pipeline, while the returns are used exclusively for realised

out-of-sample evaluation.

H.2.2 Structure Layer: Correlation–Confidence Graph

For each rolling window, cross-asset dependence is encoded in a dynamically updated graph. Pairwise correlations are transformed using Fisher's z-transform Fisher, 1915, confidence intervals are computed, and edges are retained only when the conservative lower bound of the correlation magnitude exceeds a fixed threshold. A maximum of K neighbours is assigned per asset, producing a directed neighbourhood structure $\mathcal{N}_i(t)$. This construction follows the design of Yip et al., 2023 but is recalibrated for the faster- moving FX dependence structure described by Lyons, 2001.

H.2.3 Operator Layer: Local GEDMD Estimation

For each asset and rolling window, a composite Hankel embedding is constructed using the asset and its statistically significant neighbours. Following Yip et al., 2023, the embedding is formed exclusively from *normalised price trajectories*. Given an embedding depth L, the delay vector for asset i is

$$h_i(t) = \left[\tilde{p}_i(t-L+1),\ \tilde{p}_i(t-L+2),\ \ldots,\ \tilde{p}_i(t)\right]^\top,$$

where

$$\tilde{p}_i(t) = \frac{p_i(t)}{p_i(t^*)},$$

and t^* denotes the final timestamp in the calibration window. This normalisation matches the implementation used in the accompanying notebook and ensures numerical stability in the delay-embedded matrices.

For each asset i, the delay vectors of all neighbours $j \in \mathcal{N}_i(t)$ are concatenated to form the composite observable $x_i(t)$. Snapshot matrices $X_1^{(i)}$ and $X_2^{(i)}$ are then constructed from consecutive composite states. A compact SVD is applied to $X_1^{(i)}$, and the local Koopman operator is estimated as

$$B_i = X_2^{(i)}\, V\, S_\lambda^{-1}\, U^\top,$$

with S_λ^{-1} introducing scaled Tikhonov regularisation Tikhonov, 1963. This operator structure follows the GEDMD formulation in Yip et al., 2023, where normalised prices;not returns,form the basis of the delay-embedded observable space.

H.2.4 Forecasting and Portfolio Layer

The most recent composite state is propagated τ steps ahead via iterative application of the local operator $B_i(t)$, producing predictions of normalised prices. These are mapped to predicted simple returns:

$$\widehat{r}_i(t+\tau) = \frac{\widehat{p}_i(t+\tau)}{p_i(t)} - 1.$$

The predicted-return matrix $\widehat{R}(t+\tau)$ is passed to portfolio rules implementing both threshold-based and cross-sectional ranking strategies. Walk-forward validation ensures strict non-anticipation, consistent with Cont, 2001. Final outputs include predicted returns, realised portfolio returns, Sharpe ratios, volatility, and drawdown metrics.

H.3 Pseudocode for the Computational Pipeline

The following pseudocode formalises the algorithms used in the implementation. Notation follows Schmid, 2010; Rowley et al., 2009; Yip et al., 2023.

H.3.1 Data Loading and Preprocessing

```
Algorithm H.1: PreprocessFXData
Input:
    Raw FX close-price table with columns [timestamp, asset_id, close]
Output:
    CLEAN_PRICES[T x N], RETURNS[T x N]

Steps:
1. Convert timestamp to datetime and sort rows.
2. Pivot table so each asset forms a column.
3. Forward-fill missing close prices; drop rows with remaining NaNs.
4. Compute simple returns:
       RETURNS[t, i] = P[t, i] / P[t-1, i] - 1
5. Return CLEAN_PRICES and RETURNS.
```

H.3.2 Correlation–Confidence Neighbourhood Graph

```
Algorithm H.2: BuildNeighbourhoods
Input:
    R_window[M x N], conf, c_thr, K
Output:
```

```
    Neighbourhood list NEI[i]

Steps:
1. Compute sample correlations CORR[i, j] from simple returns.
2. For each correlation:
       a. Apply Fisher-z transform.
       b. Compute z-confidence interval.
       c. Map to rho-confidence interval.
       d. Compute conservative lower bound on |rho|.
3. For each asset i:
       a. Collect all j where lower_bound(|rho_ij|) > c_thr.
       b. Always include i itself.
       c. Keep only K neighbours with largest |rho_ij|.
4. Return NEI.
```

H.2.3 Operator Layer: Local GEDMD Estimation

For each asset and window, a composite Hankel embedding is formed using the asset and its neighbours. Given embedding depth L, the composite delay vector stacks lagged *normalised prices* for asset i and its neighbours,

$$\tilde{p}_i(t) = \frac{p_i(t)}{p_i(t^*)},$$

where t^* is the final timestamp of the calibration window. Snapshot matrices $X_1^{(i)}$ and $X_2^{(i)}$ are constructed, and a compact SVD is applied. The local Koopman operator is estimated as

$$B_i = X_2^{(i)} V S_\lambda^{-1} U^\top,$$

with regularisation introduced through scaled Tikhonov ridge penalties Tikhonov, 1963. This procedure stabilises the ill-conditioned least-squares problems inherent in FX delay embeddings and follows the operator-learning structure of Yip et al., 2023 and the regularised DMD implementations in finance studied by Mann and Kutz, 2016.

H.3.4 Local Operator Estimation

```
Algorithm H.4: FitLocalOperator
Input:
    X1, X2, gamma
Output:
    Operator B_i
```

```
Steps:
1. Compute compact SVD: X1 = U S V^T.
2. Let lambda = gamma * (s_1)^2 \hfill // Scenario~3 scaled Tikhonov; Scena

3. Construct diagonal S_lambda^{-1}[k] = s_k / (s_k^2 + lambda).
4. Compute B_i = X2 * V * S_lambda^{-1} * U^T.
5. Return B_i.
```

H.3.5 Walk-Forward Forecasting Loop

```
Algorithm H.5: WalkForwardForecasting
Input:
    CLEAN_PRICES, RETURNS, (M, L, TAU),
    graph parameters, gamma
Output:
    PRED[T x N]

Steps:
1. Initialise PRED as NaN matrix.
2. For t = M to T - TAU:
       a. Define R_window = RETURNS[t-M+1 : t].
       b. Build neighbourhoods NEI via Algorithm H.2.
       c. For each asset i:
             i.   Construct X1, X2 via Algorithm H.3.
             ii.  Fit operator B_i via Algorithm H.4.
             iii. Extract latest composite state x_i(t).
             iv.  Compute TAU-step forecast of normalised price.
             v.   Convert to predicted simple return:
                     PRED[t+TAU, i] = p_hat / p_current - 1.
3. Return PRED.
```

H.3.6 Portfolio Construction

```
Algorithm H.6: ThresholdPortfolio
Input:
    PRED, RETURNS, threshold theta
Output:
    R_port
```

```
Steps:
1. For each date t with predictions:
       a. Compute directional signals s_i(t).
       b. Normalise weights w_i(t).
       c. Compute portfolio return:
             R_port[t] = sum_i w_i(t) * RETURNS[t, i].
2. Return R_port.
```